



BACHELOR THESIS

PHANTOM DATA CENTER

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ABSTRACT

Bachelor thesis
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Studio Kurilla– Prokop
14.05.26

The **Phantom Data Center** is a **speculative restoration project** of the abandoned building called the **Desfoursky Palace** in Prague to accommodate a **Storage Data Center and mixed-use programs, such as a café, a sauna and a production of dry food.**

The project investigates on the growing field of data center and the lack of space available for it by exploring the coexistence between protected architecture and contemporary digital infrastructure. This project questions **the future spatial presence of data centers within the city, their social impact and the effect it could have on rehabilitated spaces.** The research is analyzing how spatially, structurally and thermally a data center could generate new forms of occupations, rather than remaining isolated from the urban fabric.

As a speculative design proposal, the project focuses on the **resilience, adaptability and synergy between those very contrasted programs and existing building.**

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Data center
Mixed-use programs

STUDY

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98

102

RESEARCH

HOW TO KEEP OR INCREASE THE CAPACITY OF STORAGE
DATA CENTER WITHOUT BUILDING MORE ?



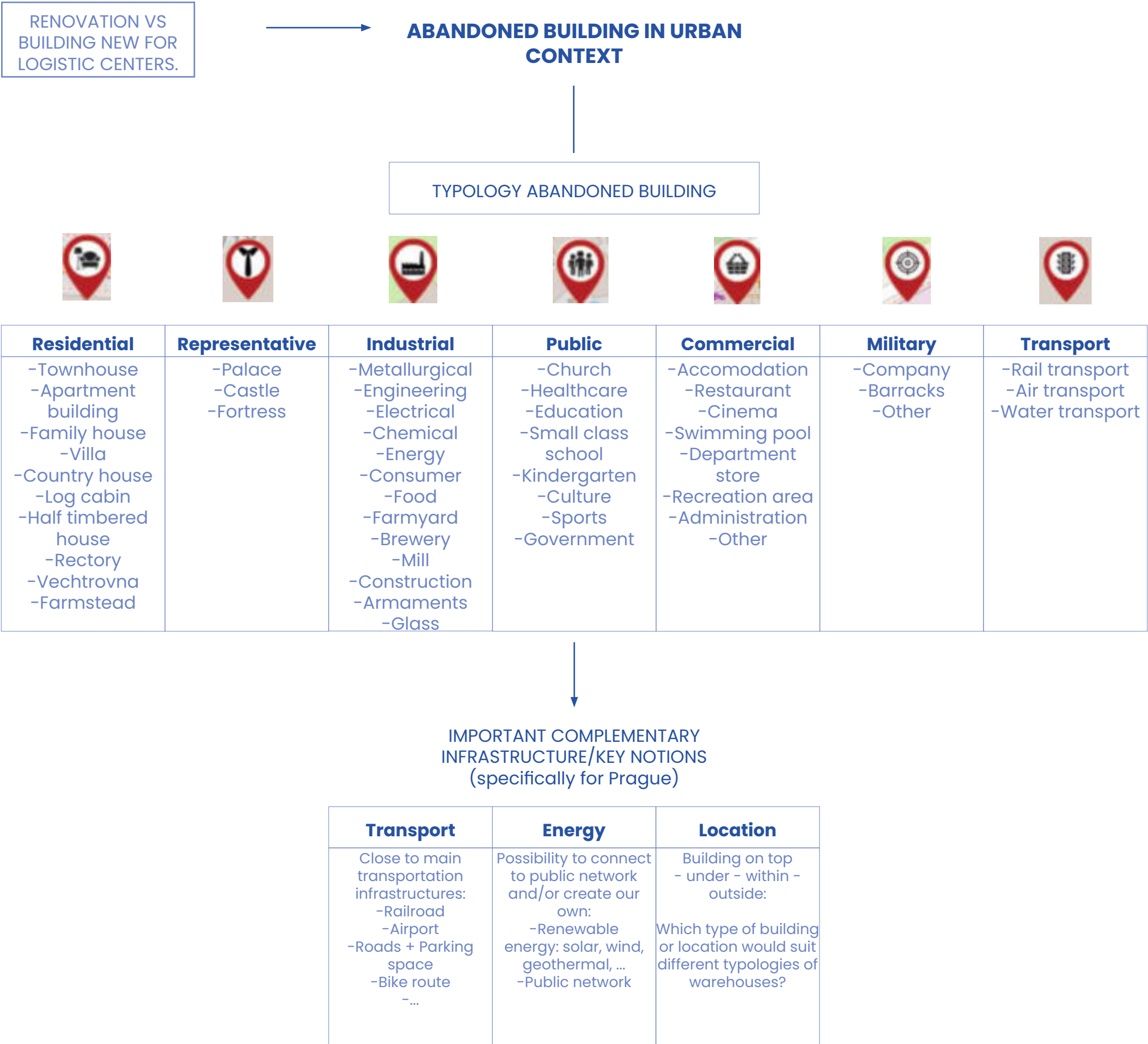
ABANDONED BUILDINGS

~ 11 billion m² of unused spaces in EU

~ 5.68 million m² in Prague

What to do with those unused spaces?

https://www.researchgate.net/publication/287575791_Prague_brownfields_Opportunity_and_threat_for_the_development_of_the_metropolis



DATA CENTER

11986 Data centers
worldwide

+ 11% growth by
2035

Where would they go?

= A facility designed to house computing, storage, and networking infrastructure. Everything held in different building typologies, capable of bearing workers and special equipments for power, cooling, shelter, and security.

Compute

= What runs your queries

- On-demand
- Temporary
- Fast



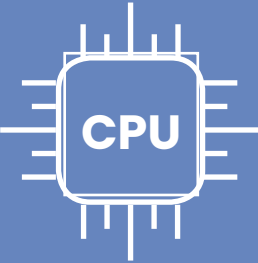
Storage

= Where your data sits

- Scalable
- Persistent
- Safe

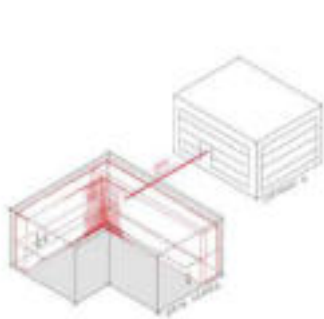
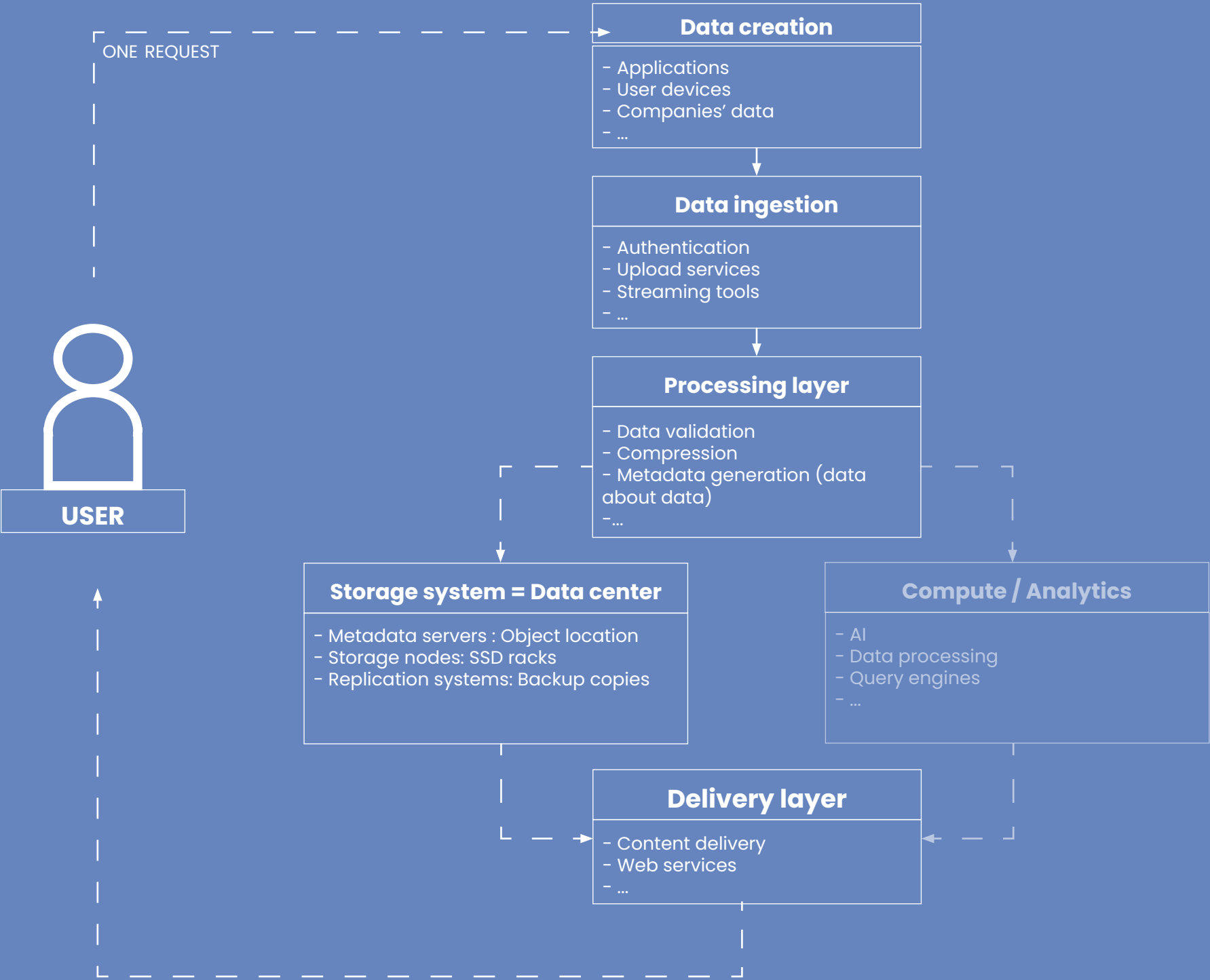


<https://medium.com/@raj.busint/storage-vs-compute-in-data-warehousing-what-it-means-and-why-it-matters-7c5c0a5bc349>

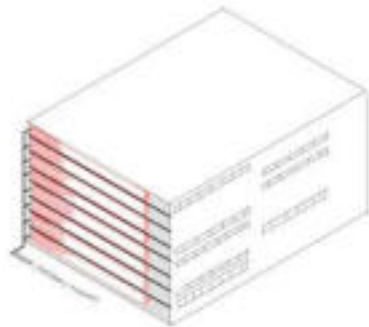


DIFFERENT TYPOLOGIES OF DATA CENTERS & FUNCTIONING

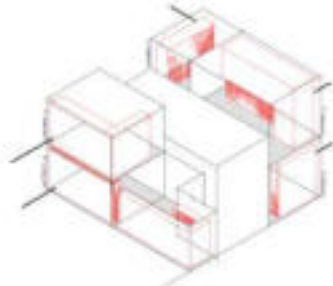
TYPOLOGIES	AREA	DESCRIPTION	MAINTENANCE NEEDS - POWER PATH	POTENTIAL POWER DENSITY	REQUIRED COOLING SYSTEM	STRUCTURAL REQUIREMENT	RETROFIT FEASIBILITY
1. ENTERPRISE DATA CENTER	4645 - 9290 m²	Support the extensive IT infrastructure of large organizations. Incorporate advanced cooling systems, high-density power infrastructure, and robust security measures.	Multiple paths for power and cooling, allowing maintenance without disrupting IT operations.	High (15-30+ kW/rack) to Extreme (50+ kW/rack)	Liquid cooling (required)	Floor load: 7 - 10 kN/m² Clear height: 3.5 - 4.5m Vibration: Low tolerance Preferred materials: Steel frame + concrete	Yes in: Former office building Telecom building Heavy industrial building
2. HYPERSCALE DATA CENTER	>9290 m²	Built and operated by big companies Support massive cloud computing and internet services. Highly optimized designs and advanced technologies like liquid cooling.	Multiple, independent, and physically isolated systems, ensuring continuous operation even during a single point of failure.	Extreme (50+ kW/rack)	Liquid cooling (exclusively)	Floor load: 10 - 15 kN/m² Clear height: 4.5 - 6m Vibration: Very strict tolerance Preferred materials: reinforced concrete frame	Very difficult but yes in: Former power plants Massive industrial structures Large distribution warehouses
3. COLOCATION DATA CENTER	465 - 930 m²	Provides a shared space for businesses. Includes on-site security, equipment, power, cooling, technical support services and the interconnection for efficient networking operations.	Multiple paths for power and cooling, allowing maintenance without disrupting IT operations.	Medium (5-15 kW/rack) to High (15-30+ kW/rack)	Hot aisle/ cold aisle containment to Liquid cooling (required)	Floor load: 8 - 12 kN/m² Clear height: 4 - 5m Vibration: Low tolerance Preferred materials: Concrete frame	Yes in: Former office building Telecom building Heavy industrial building
4. EDGE DATA CENTER	<465m²	Located closer to end-users to reduce latency. Caching content or processing data. House a few dozen to a few hundred servers.	Single path from utility source to IT equipment.	Low (2-5 kW/rack) to Medium (5-15 kW/rack)	Air cooling to Hot aisle/ cold aisle containment	Floor load: 5 - 8 kN/m² Clear height: 3 - 4m Vibration: Medium tolerance Preferred materials: Reinforced concrete frame	Yes in: Basement Parking garages Industrial buildings
5. MODULAR DATA CENTER	<465m²	Portable data centers Modules contain IT equipment for power and cooling.	Single path from utility source to IT equipment.	Low (2-5 kW/rack) to Medium (5-15 kW/rack)	Air cooling to Hot aisle/ cold aisle containment	Floor load: 5 - 10 kN/m² Clear height: 3- 5m Vibration: Medium tolerance Preferred materials: Steel frame	Excellent



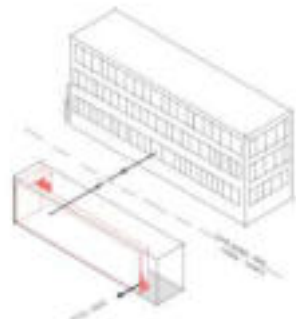
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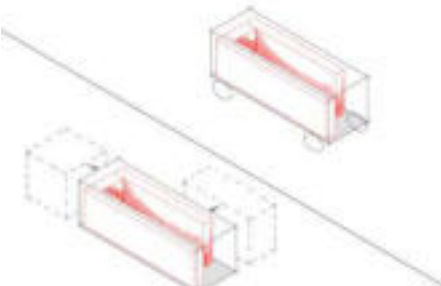
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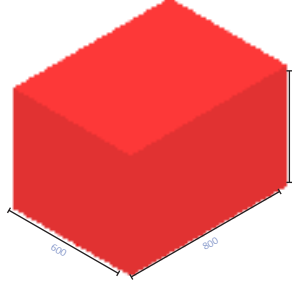
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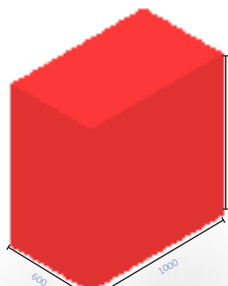
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5



12U



24U



42U

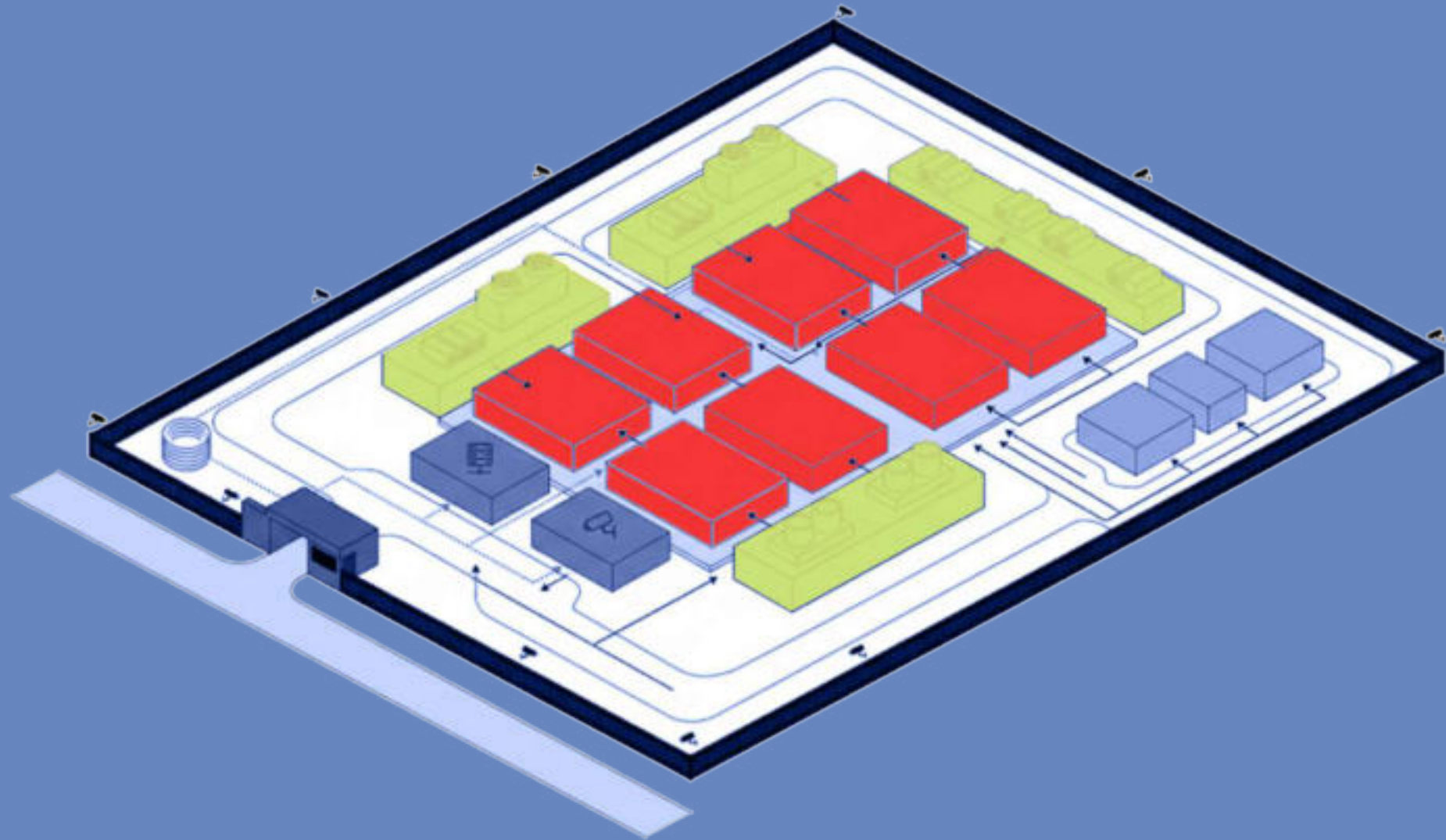


45U



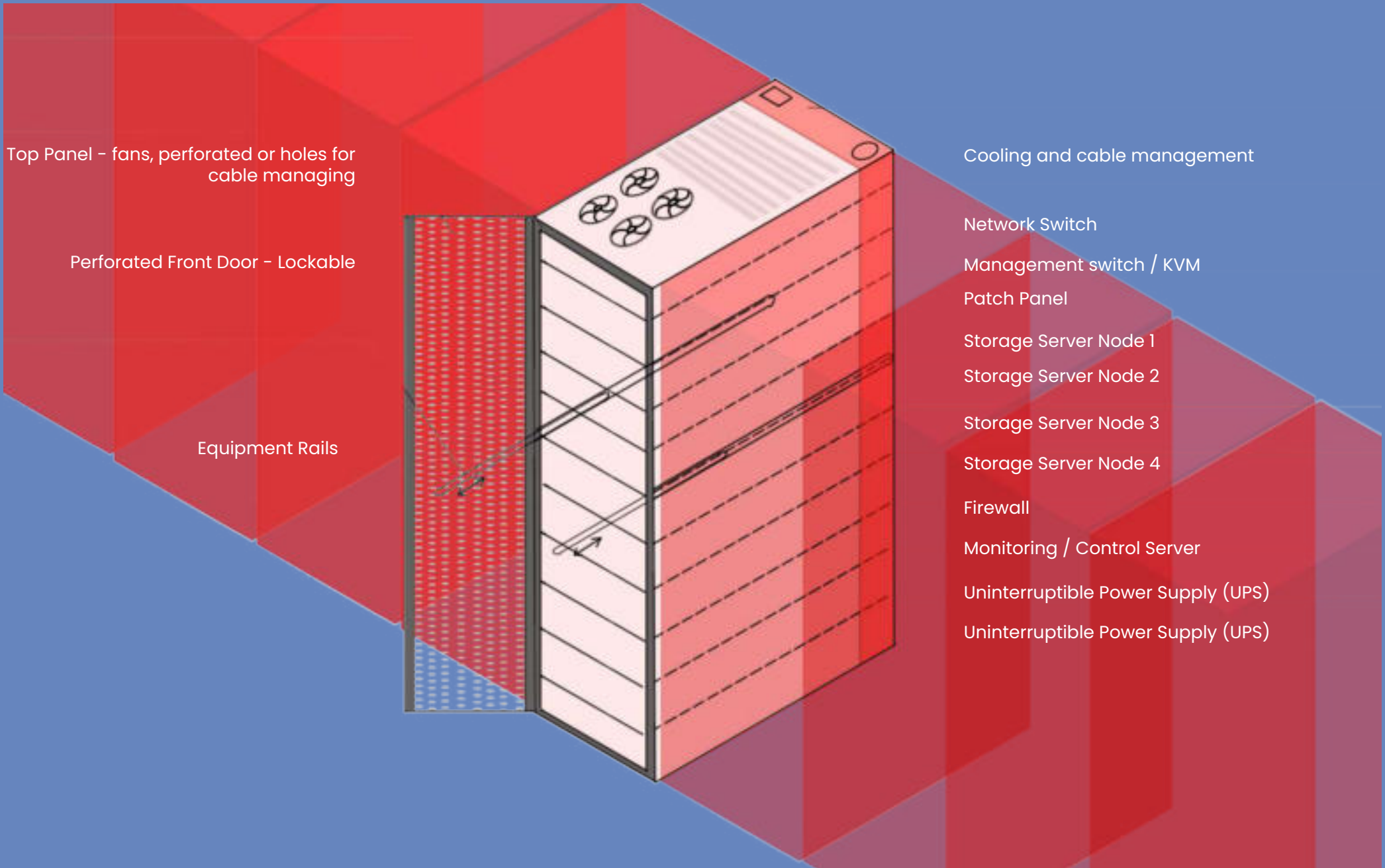
48U

COMPONENTS OF A REGULAR DATA CENTER



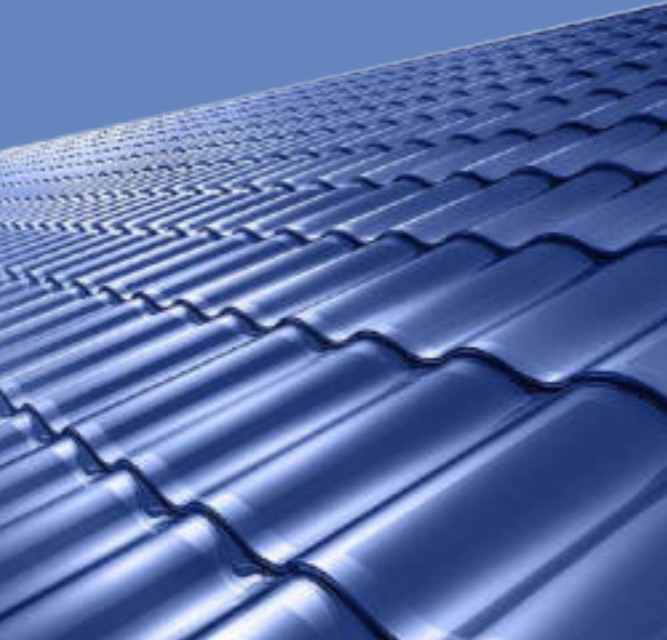
- Racks infrastructure
- Cooling infrastructure

COMPONENTS OF A RACK



INFRASTRUCTURES NEEDED FOR ENERGY – HEATING RECOVERY – COOLING:

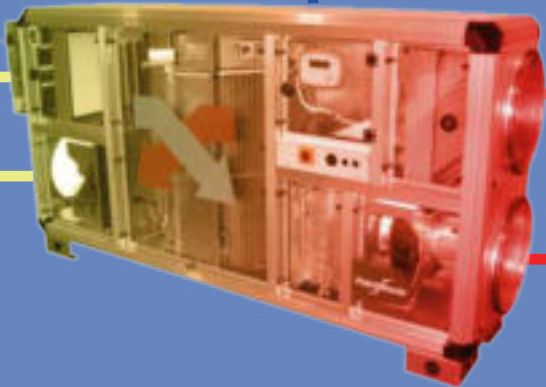
SIZE SERVER	VOLUME (m3)	POWER DENSITY (kW/rack)	TEMPERATURE RANGE NEEDED (°C)
12 U	0.26	Low: 2-5 kW/rack	28-35
24 U	0.64	Medium: 5-15 kW/rack	30-40
42 U	1.12	Medium: 5-15 kW/rack	35-45
45 U	1.44	High: 15-30 kW/rack	40-50
48 U	1.54	Extreme: over 50 kW/rack	45-60



Solar tiles

Energy

MIXED-USE
PROGRAMS



Heat exchanger

Heating recovery



Direct to chip liquid cooling



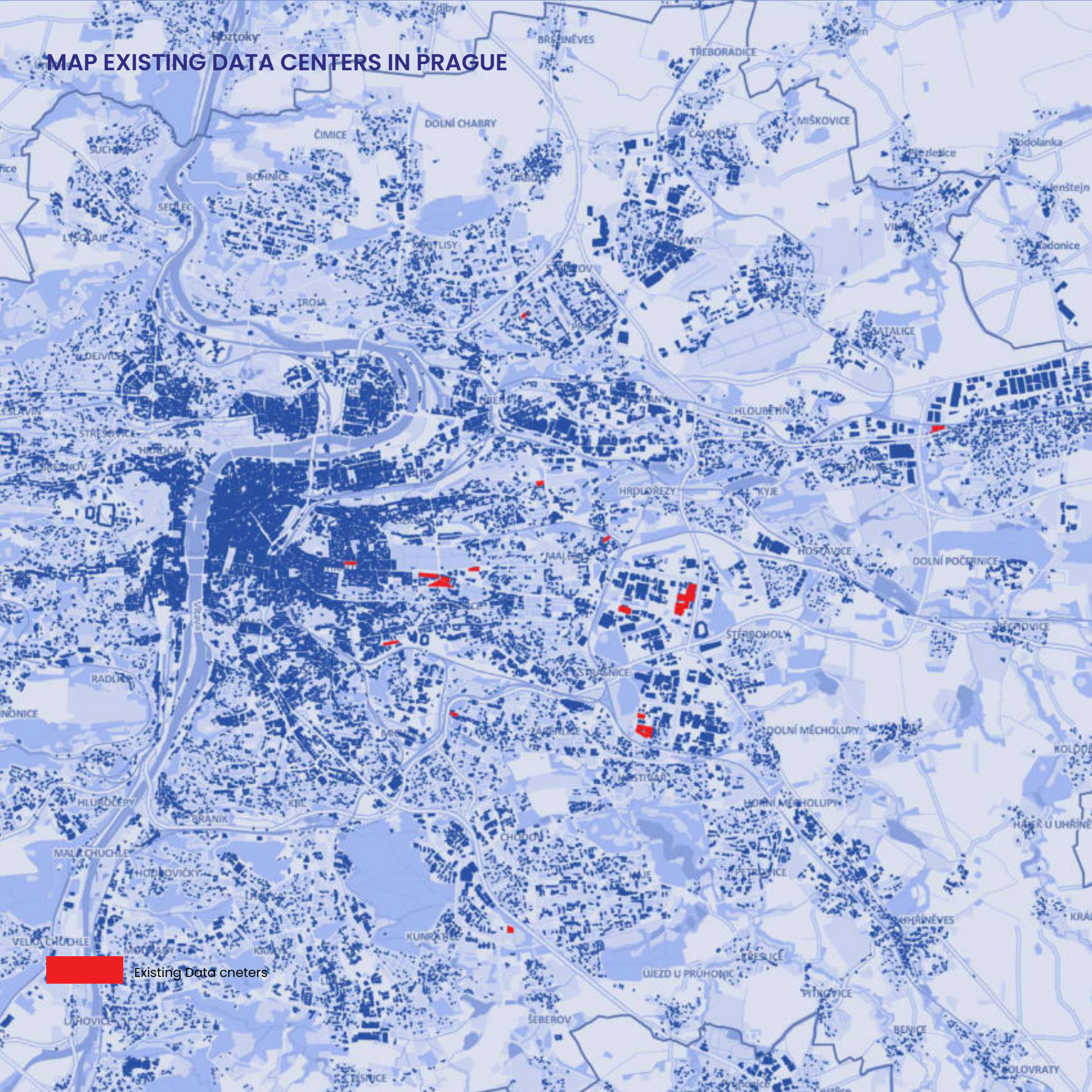
Fan



Dry cooler

Cooling





MAP EXISTING DATA CENTERS IN PRAGUE

WHY DATA CENTER?

SCALABILITY

The components of a Data Center is helping to make the organization somewhat modular. The gradual and flexible capacity of racks can be changed depending on the demand for storage. Also due the maintenance needs of a Data Center, the amount of IT members is relatively low, making programs necessary for people lower. Something that isn't allowed from formal logistic infrastructures requiring large open storage volumes and intensive transportation systems.

URBAN INTEGRATION

Compared to conventional logistics infrastructures a Data Centers are less of an eyesore, with potential noise pollution and transportation flows. Data Center can then cohabit within the city thanks to their scalability.

HEATING BENEFIT FOR SURROUNDINGS

The racks from a Data Center are creating continuous heat that is usually a waste rather than turning it into something beneficial for the architectural, and/or urban scale. Which could improve the idea behind a circular economy and synergy with the surroundings.

LISTED ABANDONED BUILDINGS IN PRAGUE – MATRIX

S



Area: 134.7m²



Area: 428.96m²



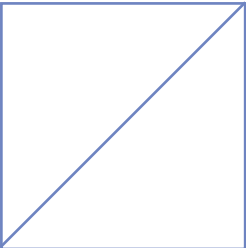
Area: 352.7m²



Area: 560.78m²



Area: 168.11m²



Area: 152.14m²

M



Area: 177.68m²



Area: 1298.03m²



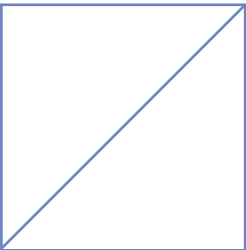
Area: 634.49m²



Area: 968.26m²



Area: 544.33m²



Area: 447.77m²

L



Area: 290.77m²



Area: 1547.44m²



Area: 946.86m²



Area: 1715.46m²



Area: 840.59m²



Area: 1345.86m²



Area: 529.66m²

XL



Area: 715.68m²



Area: 2613.72m²



Area: 2855.95m²



Area: 19115.46m²



Area: 1240.2m²



Area: 2696.23m²



Area: 4497.75m²



Commercial – S



Representative – M



Transport – L



Public – XL

		Location																									
		Residential				Representative				Industrial				Public				Commercial				Military		Transport			
		S	M	L	XL	S	M	L	XL	S	M	L	XL	S	M	L	XL	S	M	L	XL	L	XL	S	M	L	XL
Typology Data Centers	Enterprise Data Center (4645 - 9290 m²)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	O	X	X	X	X	X	X	X	X	X	X
	Hyperscale Data Center (>9290 m²)	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	O	X	X	X	X	X	X	X	X	X	X
	Colocation Data Center (465 - 930 m²)	X	X	X	M	X	O	O	O	X	M	O	O	M	O	O	O	X	M	M	O	O	O	X	X	M	O
	Edge Data Center (<465m²)	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	M	O	O	O	O	O	M	M	O	O
	Modular Data Center (<465m²)	M	M	O	O	O	O	O	O	O	O	O	O	O	O	O	O	M	O	O	O	O	O	M	M	O	O

X Data center typology size **doesn't match** with location
O Data center typology size **does match** with location
M Data center typology size **could match** with location



THE DESFOURSKY PALACE

Na Florenci 21a, 110 00 Nové Město

The Desfoursky Palace in Florenc, Prague's New Town, is used as a study case for this project. Built by the architect [Josef Kranner in 1847](#), the Palace was owned by the Franz Desfours-Walderode. In 1878 the Palace got sold by his wife.

In the second half of the 20th century, the palace of the daily Red Law served, which was greatly reflected in its interiors.

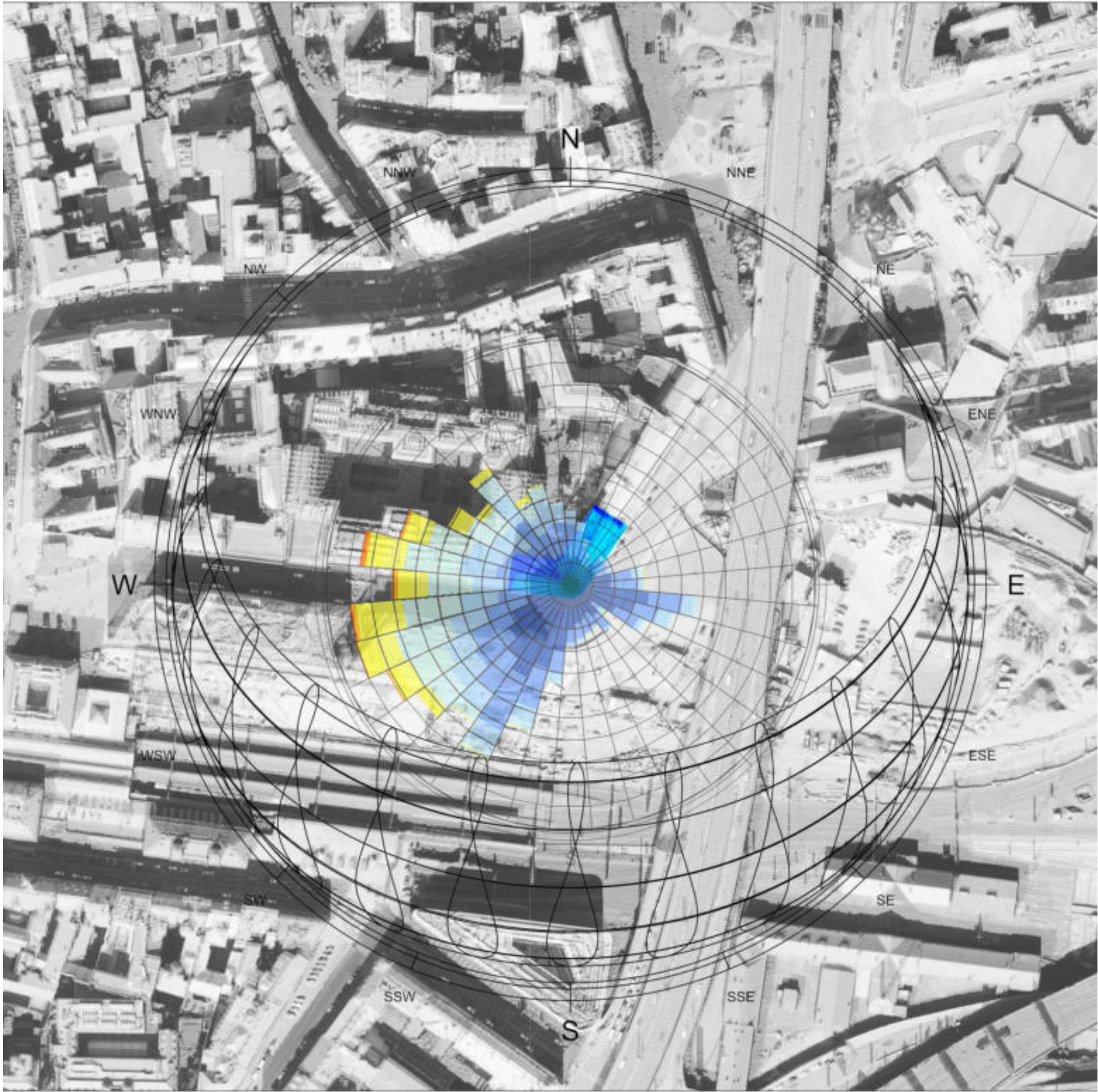
One of the youngest noble palaces in Prague also belonged to the Krasnopolski family for several decades, whose last members were murdered by the Nazis.

Ever since then, the building had no official owner and was considered to be demolished. The ownership was transferred to the city of Prague, without any ambition for it.

1980: Listed as protected cultural monument by the City of Prague.







1:2000



1:500

	Heat recovery		Energy consumption	Water consumption	Flexibility/ Adaptability / Scalability	Community & Social impact
	Winter	Summer				
How to calculate them?	Amount of heat available to recover and the area possible to cover (kW)		Amount of energy used and needed for data centers compared with other type of energy. (kW)	Volume of water used and needed for data centers (m3)	Volume range of the racks and so the data center structure. (m3)	Amount of job created. (job/year)
Mix-Use programs						
Cultural/ Event space	10-30 kW Intermittent heating	5-15 kW Not constant demand	5-15 kW	50-100	5000-15000 m3 High flexibility	?
Workshops	40-80 kW Process and ambient heating	30-70 kW Depending on activity	15-40 kW	100-300	4000-10000 m3 High flexibility	?
Sauna/ Wellness Space	120-200 kW Constant demand	100-180 kW Strong demand	40-120 kW	500-1500	2000-6000 m3 Moderate flexibility	20-40 jobs
Laundromat	100-150 kW Constant hot water need	100-150 kW Same in summer	50-100 kW	1000-3000	2000-5000 m3 Moderate flexibility	?
Cafe	30-60 kW Moderate demand	10-30 kW Internal heat gains dominate	20-60 kW	200-600	1000-3000m3 Moderate flexibility	10-20 jobs
Bakery	30-60 kW Limited additional heating need	10-20 kW Overheating risk	30-80 kW	100-300	1000-3000 m3 Low flexibility	?
Brewery	80-140 kW High demand for process heating	80-140 kW Constant demand	50-120 kW	2000-6000	3000-8000 m3 Low flexibility	?
Dry food production	120-180 kW Continuous drying process	120-180 kW Constant demand	60-150 kW	200-800 m3	3000-7000 m3 Low flexibility	50 jobs due to the low automatisisation system
Research labs	60-120 kW Controlled environment	60-120 kW Constant demand	80-200 kW	300-1000	2000-6000 m3 Low flexibility	?
Controlled Vertical farming	150-250 kW Climate control	150-250 kW Constant demand	100-300 kW	100-300 ??	4000-9000 m3 Moderate flexibility	?

MIXED-USE PROGRAMS

CAFE

Although the cafe doesn’t require a significant thermal energy, it is interesting concerning the social impact of the building. It represents a flexible and accessible public program that can be easily integrated into a rapidly evolving urban district.

SAUNA/WELLNESS SPACE

This program is connected here to the Data Center for its constant demand of heat all year long. The possible relationship between a direct continuous heat source and the need for a controlled environment emphasis the connection of digital infrastructure and human occupation. The placement in a very touristic area makes this type of program the perfect choice.

DRY FRUITS PRODUCTION

The stable and hygienic conditions needed by a dry food production, fits the temperature-controlled and heat production from the Data Center . The constant need on heat for the drying process while providing the neighborhood a new form of economy, improve the overall social impact. Acting as a small business to provide to the area local dry fruit, grains and granola for any type of usage.



● Concept

Area : 1298.03m²



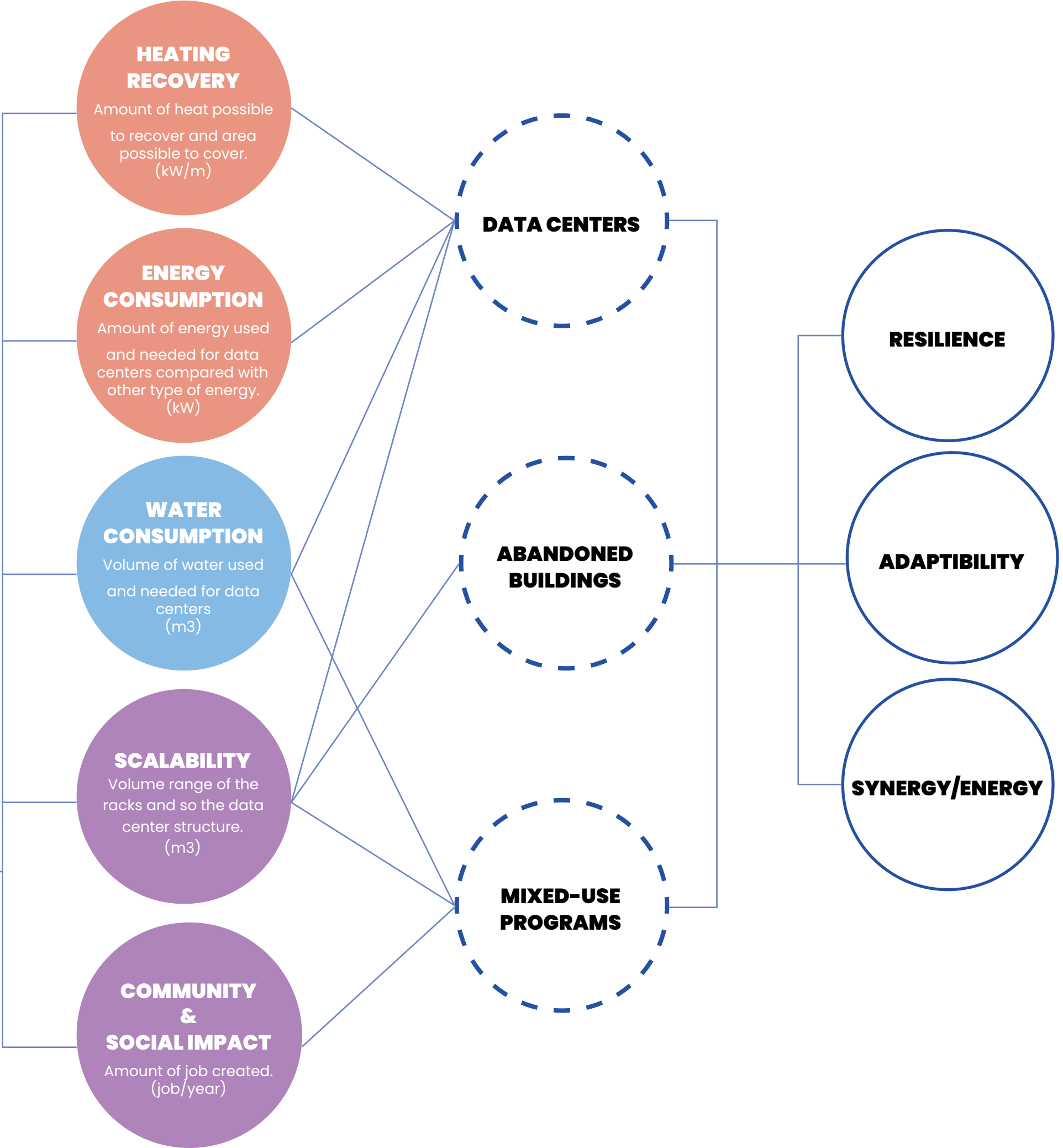
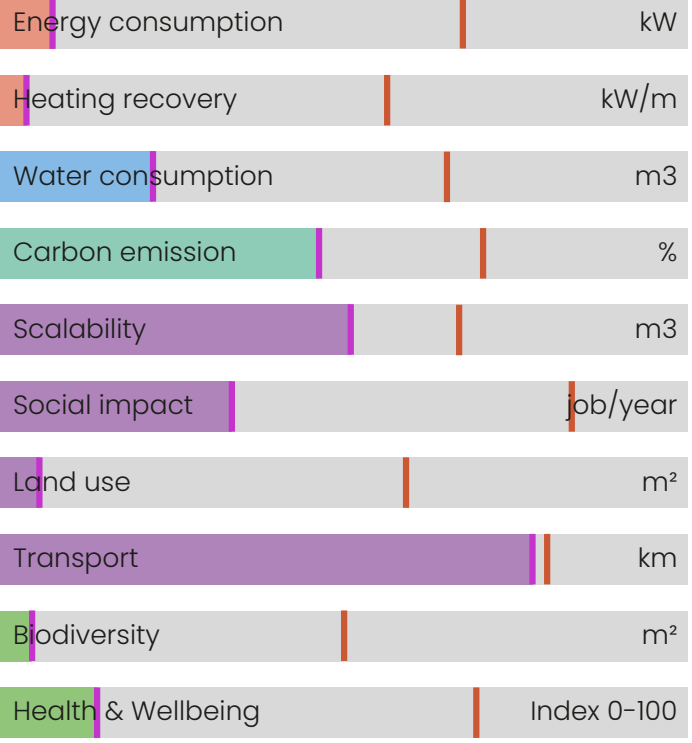
● Proposal

Area : 1298.03m²

Description and, or research questions

- Amount of energy used and needed for data centers compared with other type of energy. Unit: kW
- Amount of heat possible to recover and area possible to cover. Unit: kW/m
- Volume of water used and needed for data centers; Unit: m3
- Amount of carbon emission depending on the sector. Unit: %
- Volume range of the racks and so the data center structure. Unit: m3
- Amount of job created. Unit: job/year
- Area ratio between building footprint and green-blue spaces. Unit: m²
- Distance and accessibility for goods and users. unit: km
- Area of green-blue spaces accessible to public, animals and insects. Unit: m²
- Indicator of health & wellbeing based on: pollution, safety and stress. Unit: Index 0-100

KPI names



STUDY

P0: BUILDING MODIFICATIONS

P1: 80% MIXED-USE 20% DATA CENTER

P2: 50% MIXED-USE 50% DATA CENTER

P3: 20% MIXED-USE 80% DATA CENTER



QUALITY ANALYSIS

- Accessibility**
 - Entrance point
 - Main entrance
- Structural integrity**
 - Removed wall
 - Kept wall
 - New structural system
- Room quality**
 - High
 - Medium
 - Low



0 5.5 11 m
2PP - 1:200

- Accessibility**
 - Entrance point
 - Main entrance
- Structural integrity**
 - Removed wall
 - Kept wall
 - New structural system
- Room quality**
 - High
 - Medium
 - Low
- View**
 - High
 - Medium
 - Low
- Natural Light Intake**
 - High
 - Medium
 - Low



0 5.5 11 m
1PP - 1:200



ARCHITECTURE
DIZAJNARSKI ILLUSTRACIJSKI
POSREDOVANJE ZA PROJEKTOVANJE

ORGANIZATOR
POSREDOVANJE ZA PROJEKTOVANJE
POSREDOVANJE ZA PROJEKTOVANJE

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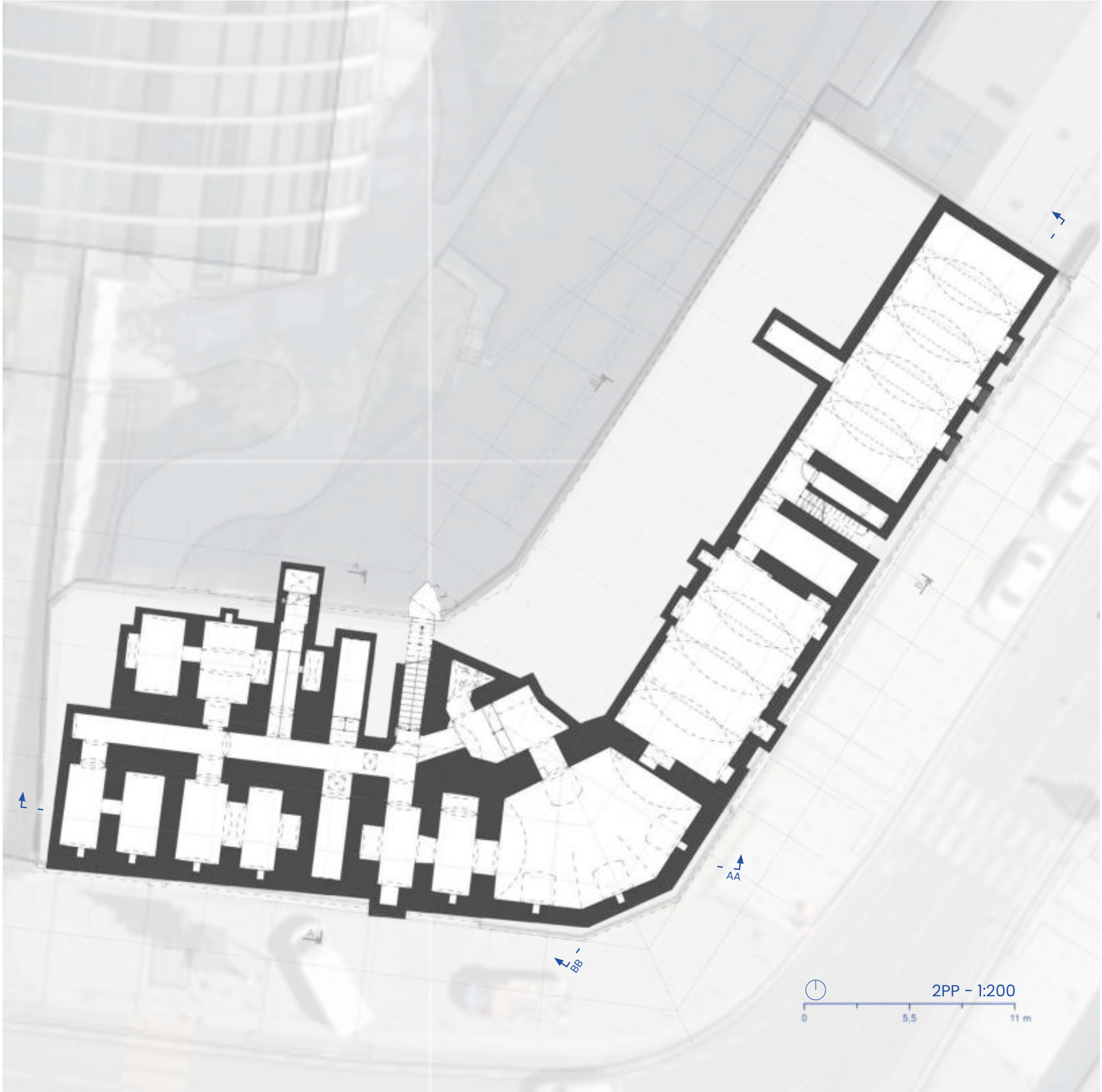


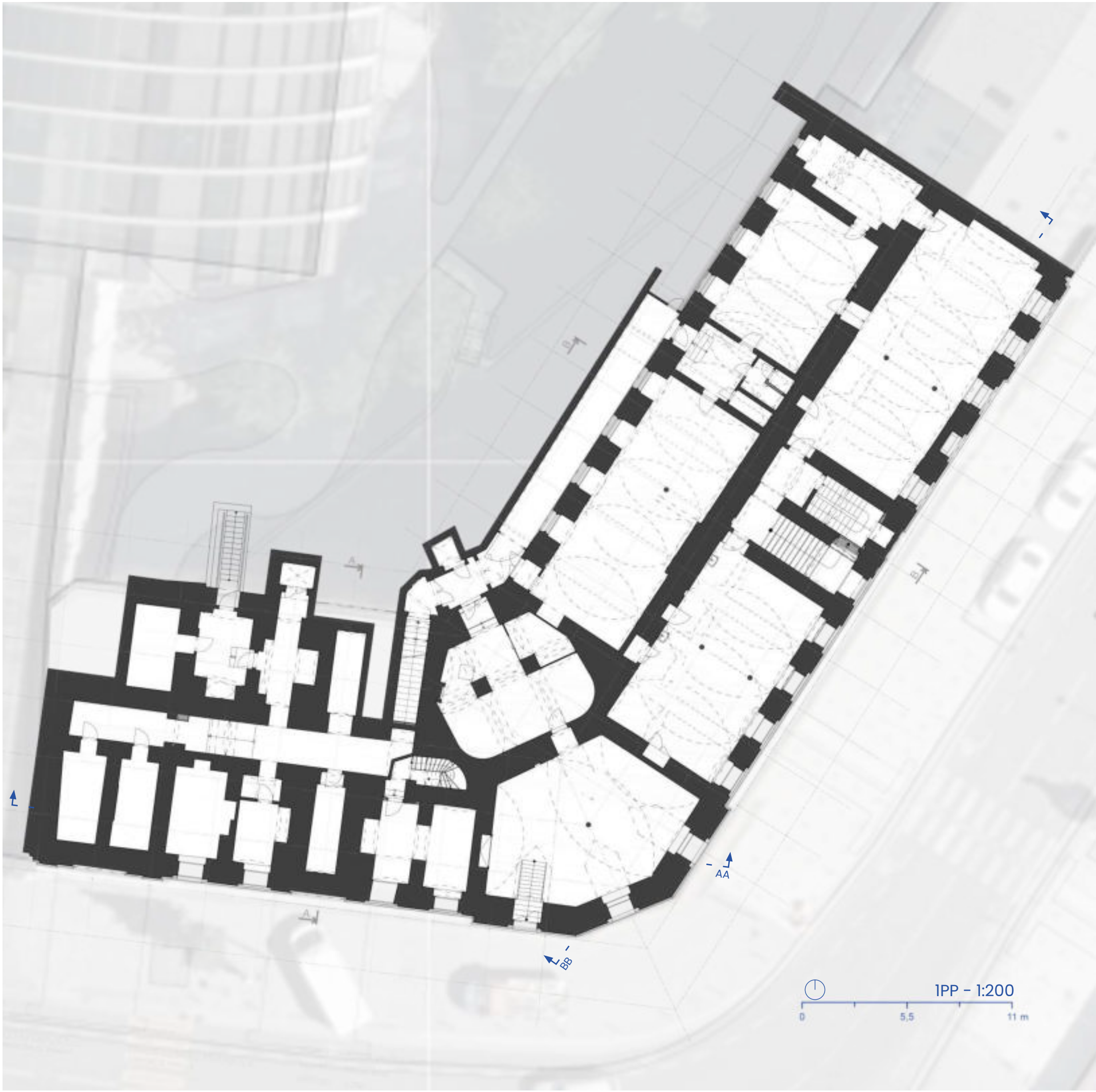
Before the evolution of each phase, a preceding phase is established to improve the interior layout to accommodate the new programs.

A grid of 3 by 3 meters is helping with the inside organization. A column and beam system is being introduced replacing the existing partitions and some of the load-bearing walls. Offering bigger spaces to allow for flexibility of programs.

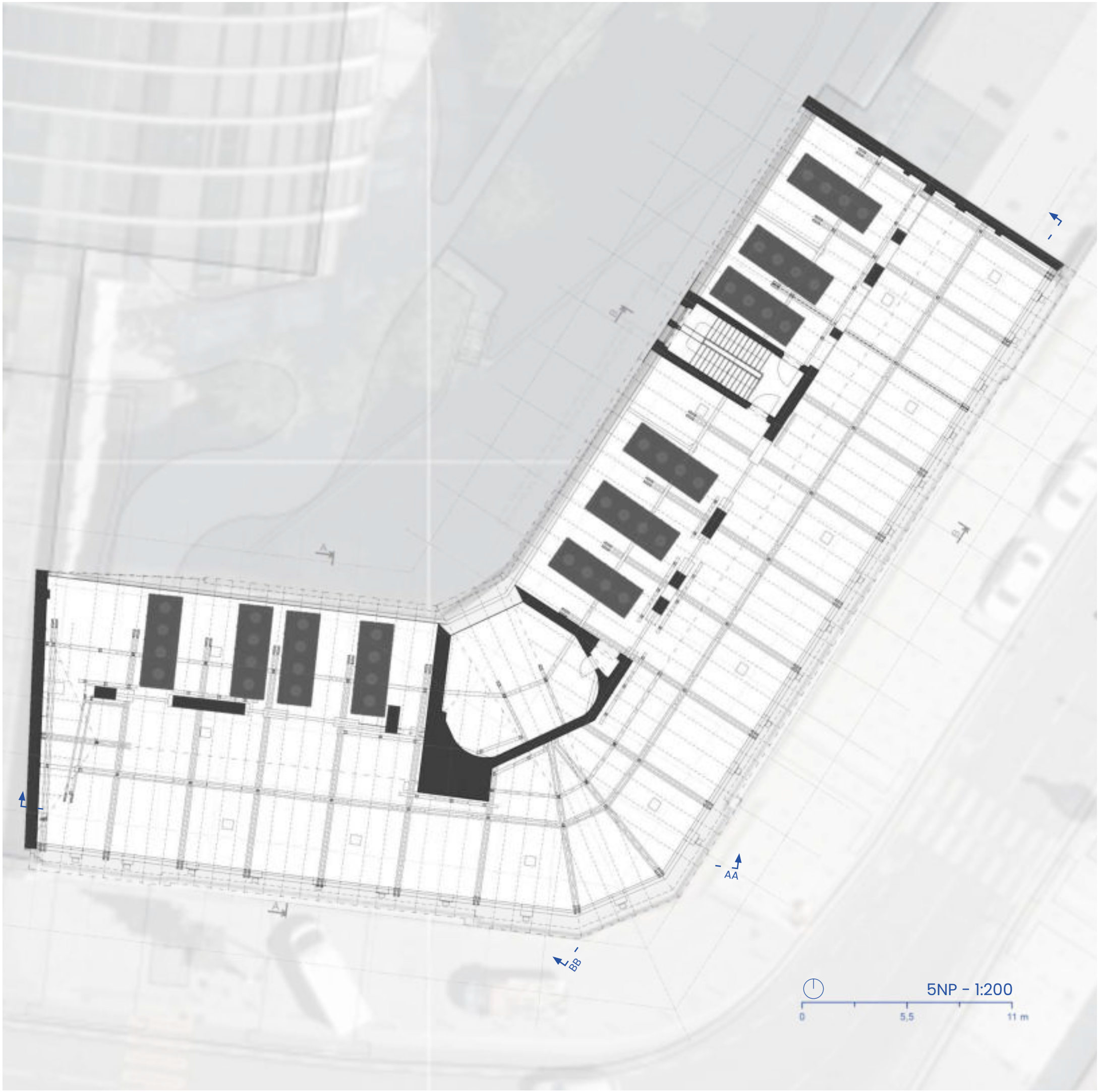
Something else that is taken care of in that phase is the HVAC system covering 100% of the building. This way even with the Storage Data Center expanding, the technical infrastructure needed is already in place. The last thing changing ,complementary to the cooling system, is the roof. The roof structure is being remodeled to form an open area for the air supply.

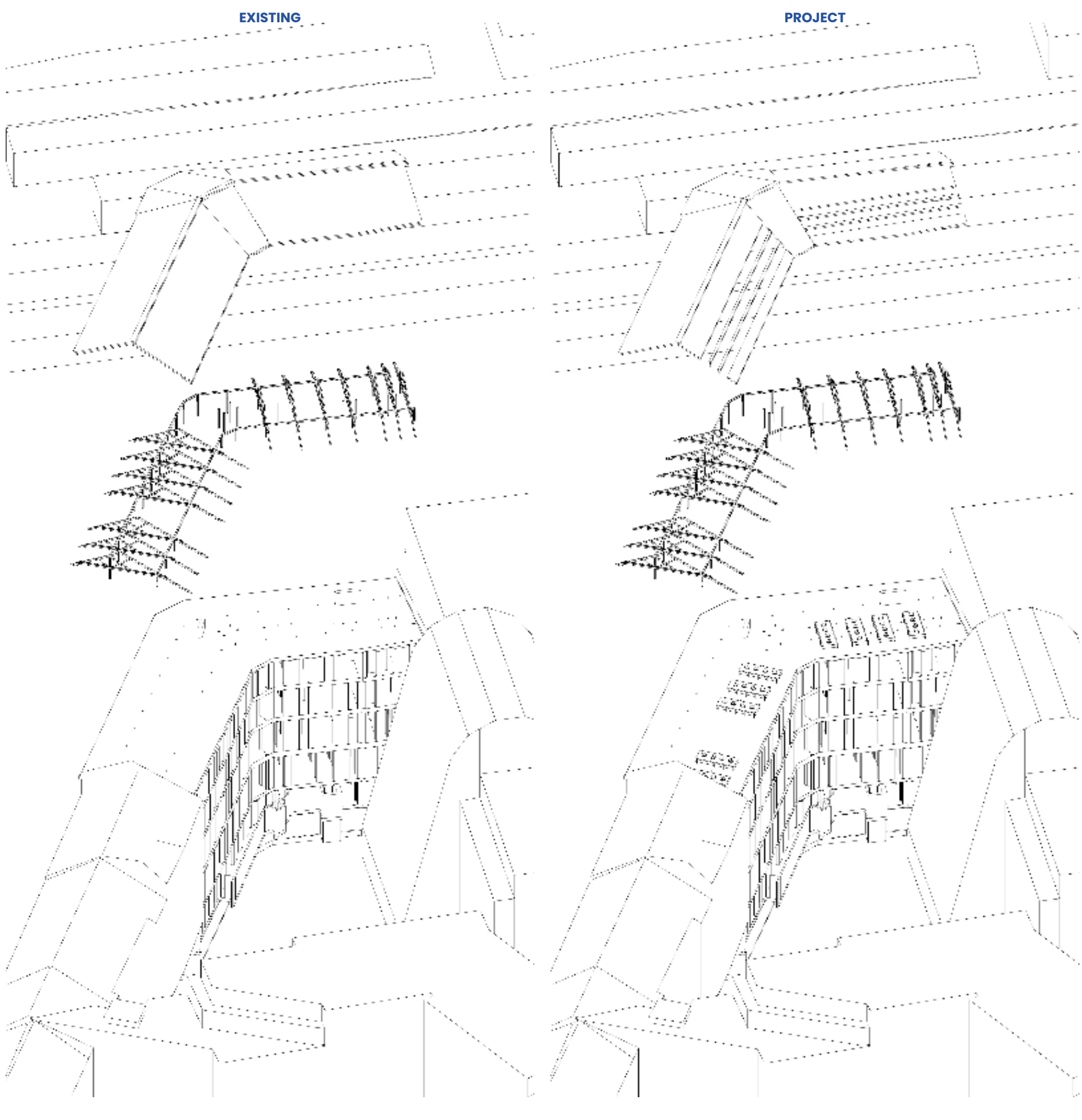
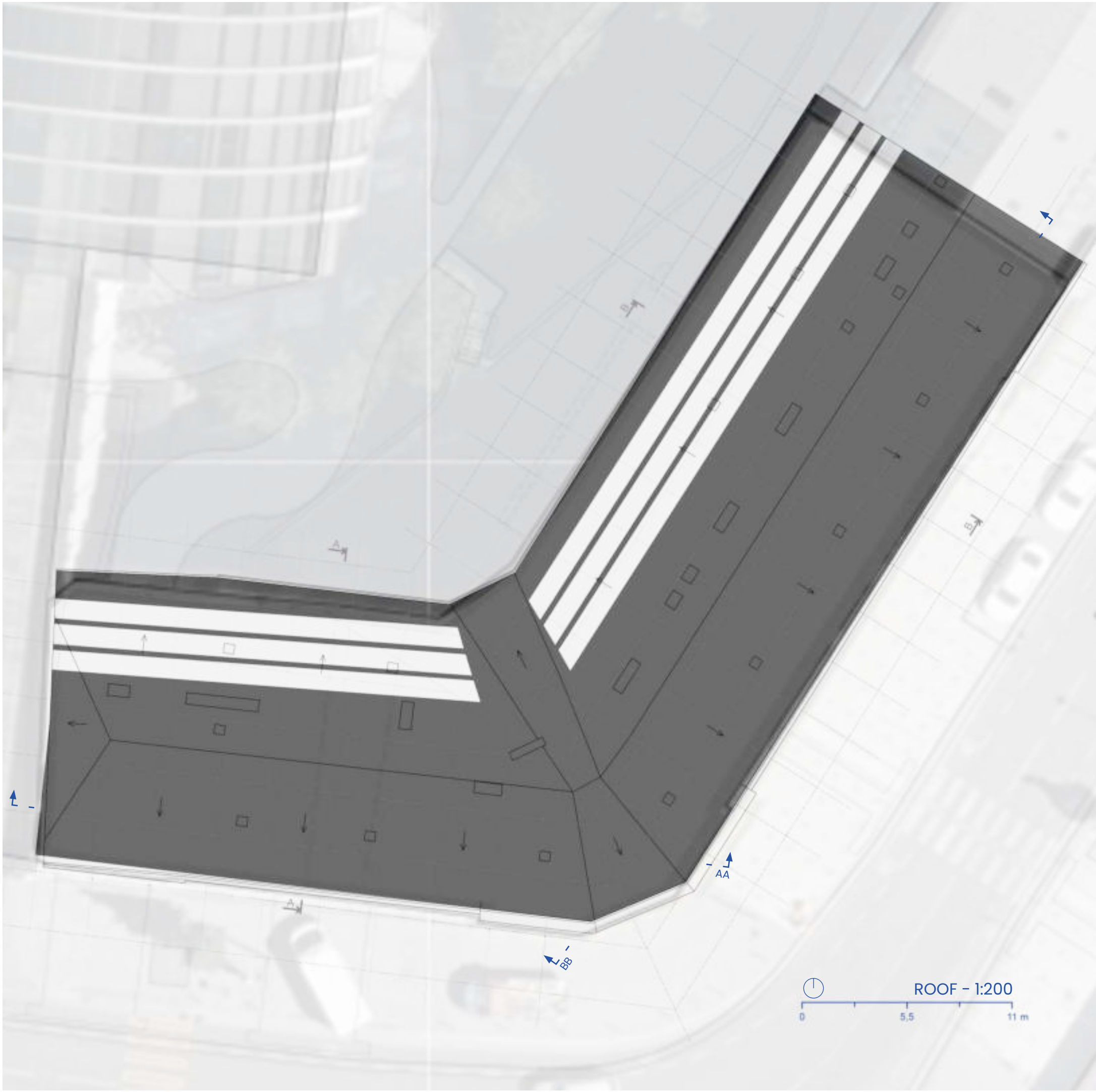
P0
ARCHITECTURAL
INTERVENTIONS

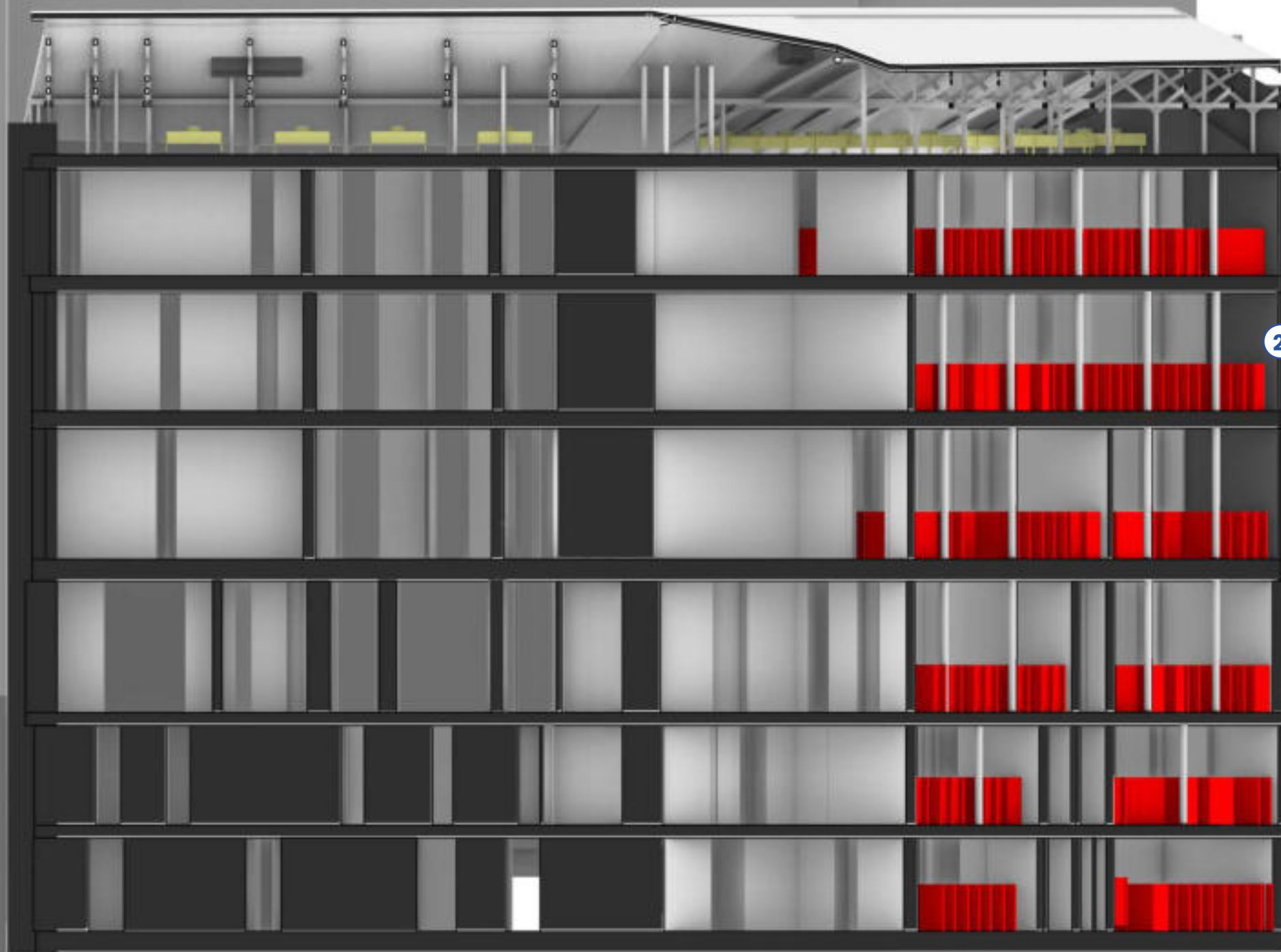




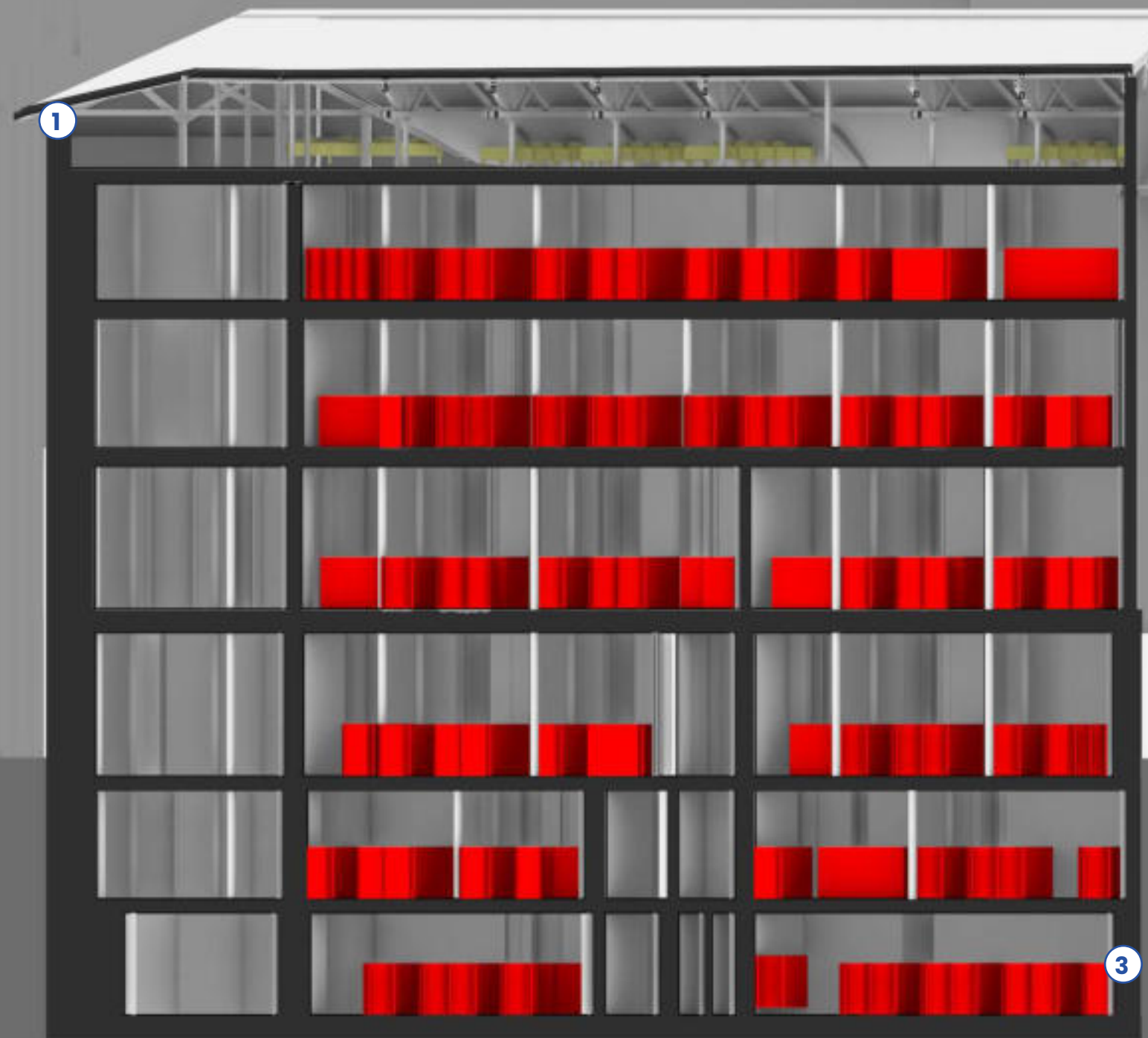








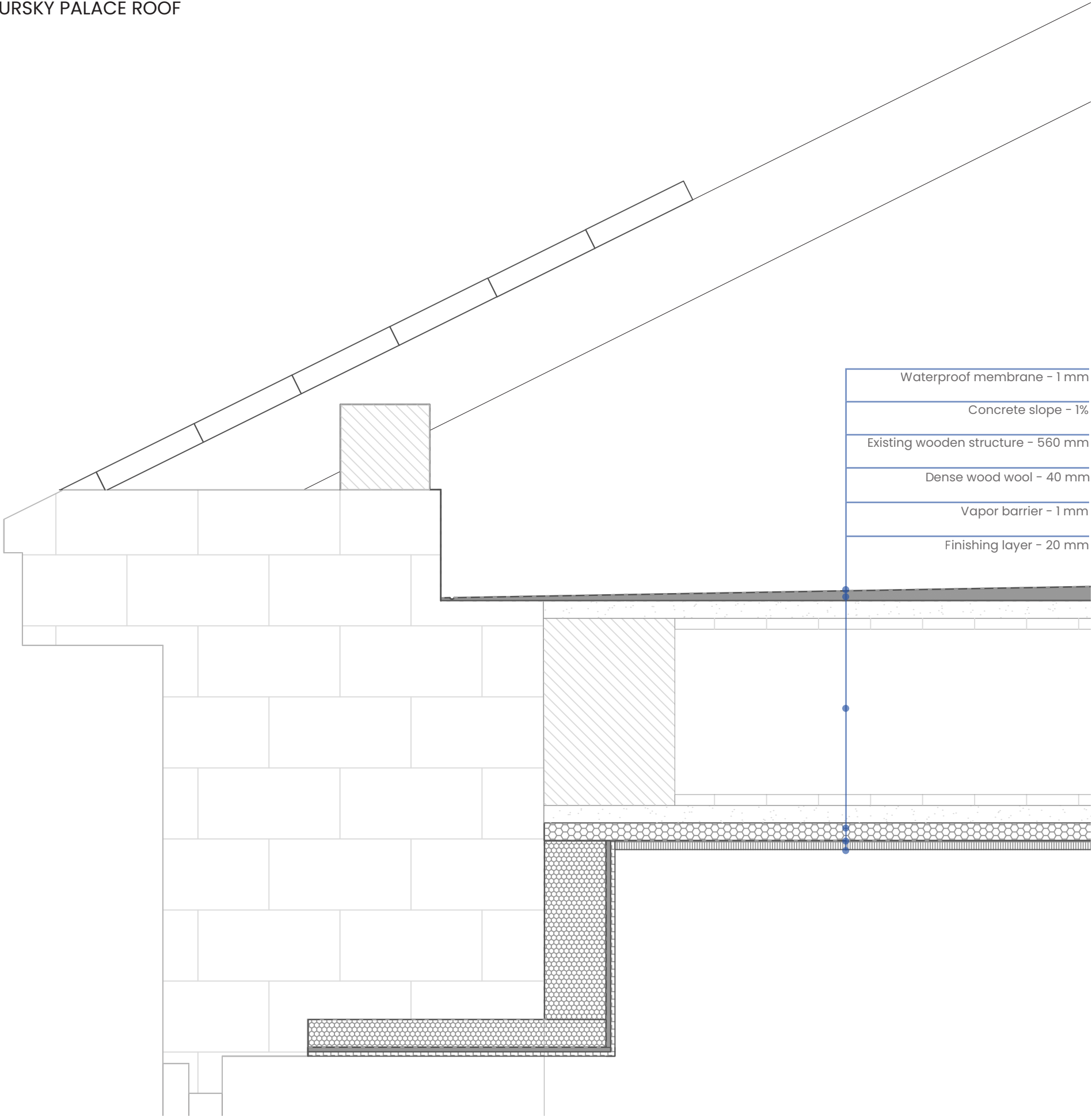
SECTION AA' - 1:200
0 5,5 11 m



SECTION BB' - 1:200
0 5,5 11 m

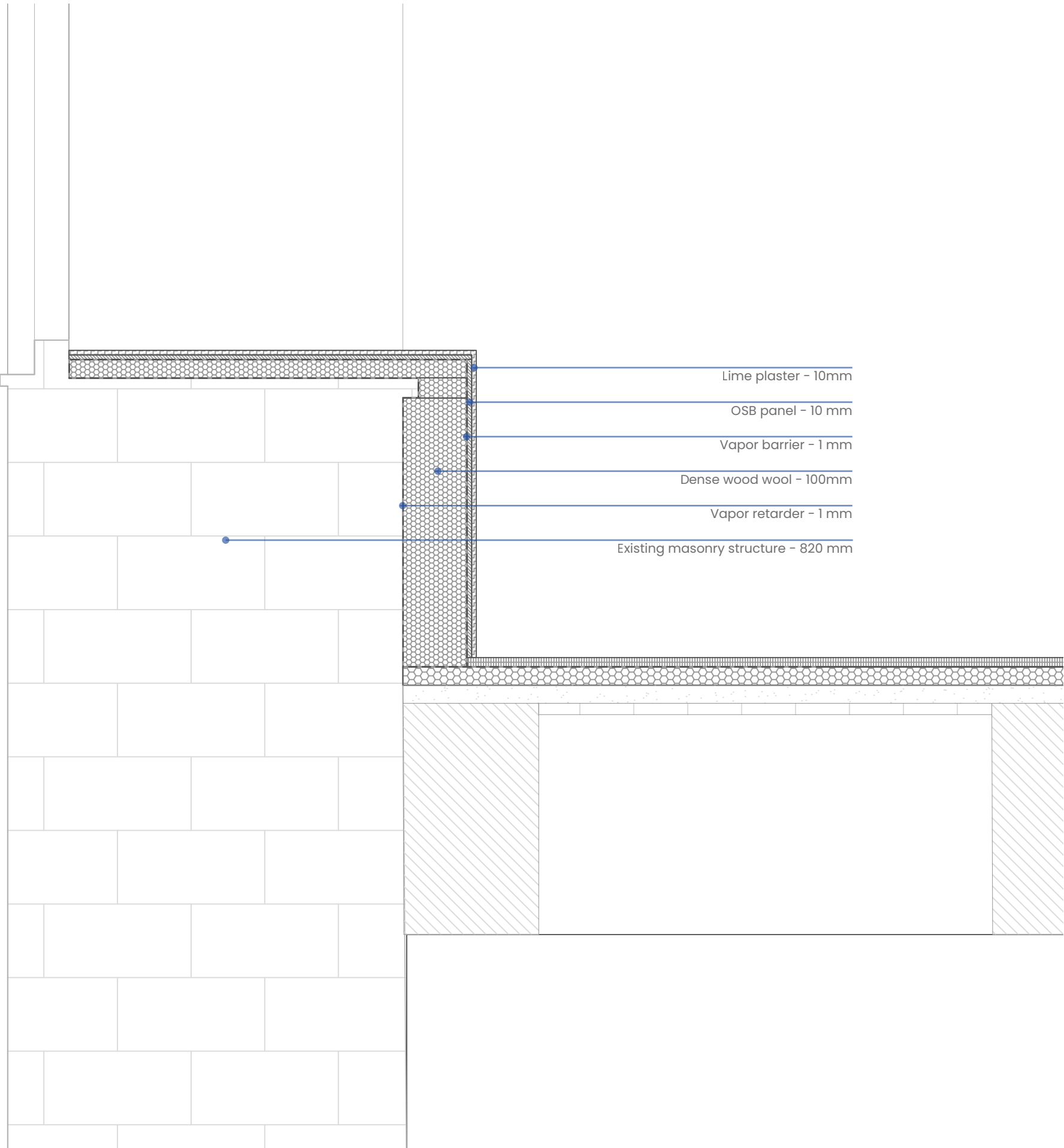
1.ROOF DETAIL

1:10 – DESFOURSKY PALACE ROOF



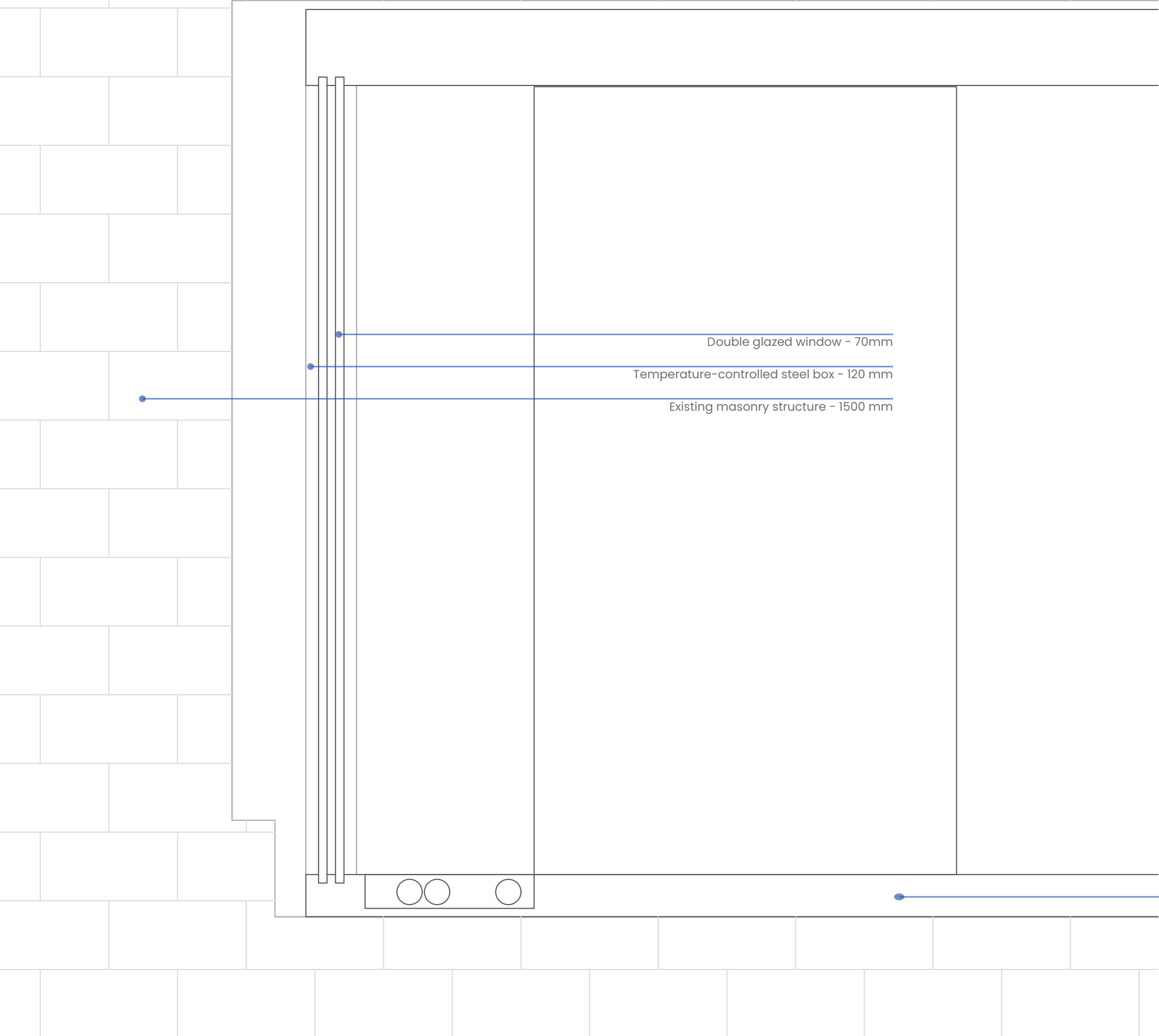
2.WALL DETAIL

1:10 – DESFOURSKY PALACE WALL

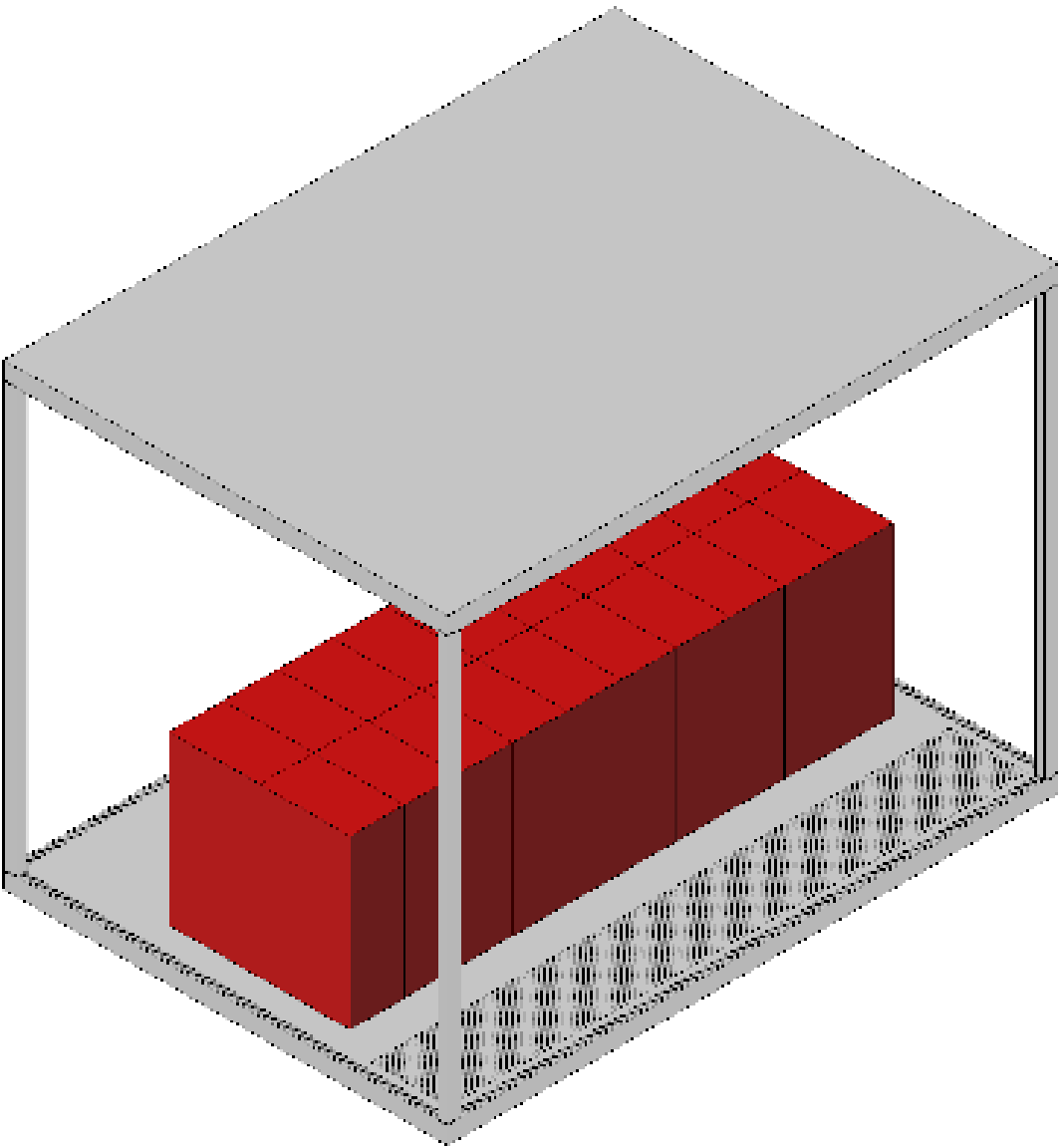


3.FOUNDATION DETAIL

1:10 – DESFOURSKY PALACE FOUNDATION



TEMPERATURE-CONTROLLED BOX



The first phase is focusing on a more realistic approach with the racks placed in the basements of the existing structure. This proposition analyze the possibility of that happening and would help solving the issue concerning the lack of space for such a growing field.

The main portion of the building in this phase is the mixed-use programs. Because of the location and the developing residential area, the picked programs are not only taking into account the heat recovery but also the potential positive social impact.

The Storage Data Center is then beneficial for the Palace and the mixed-use programs implemented.

P1
80% MIXED-USE
20% DATA CENTER



MIXED-USE PROGRAMS

DRY FRUITS PRODUCTION

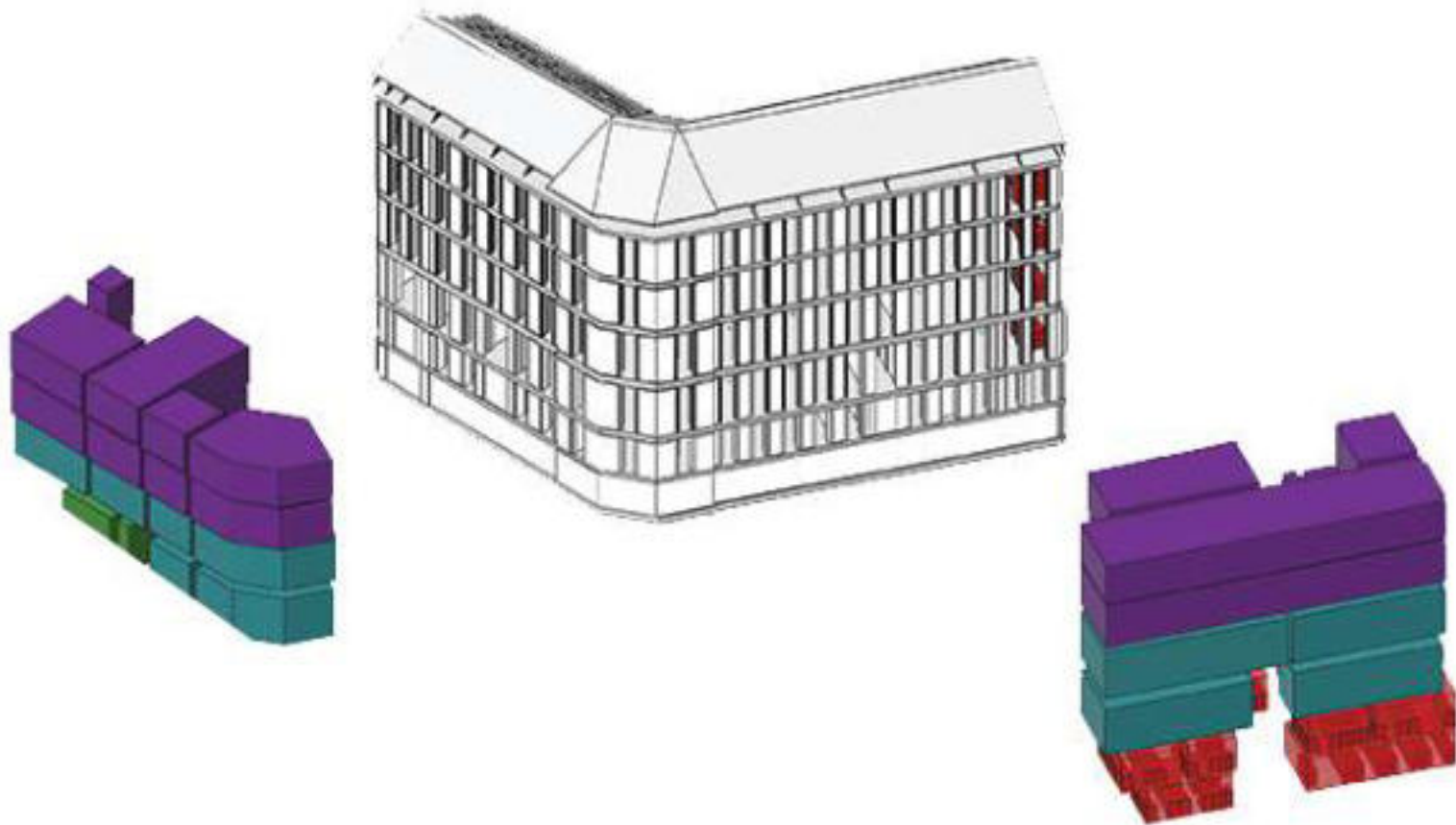
P1 5293 m²

SAUNA/WELLNESS SPACE

P1 4760 m²

CAFE

P1 587 m²



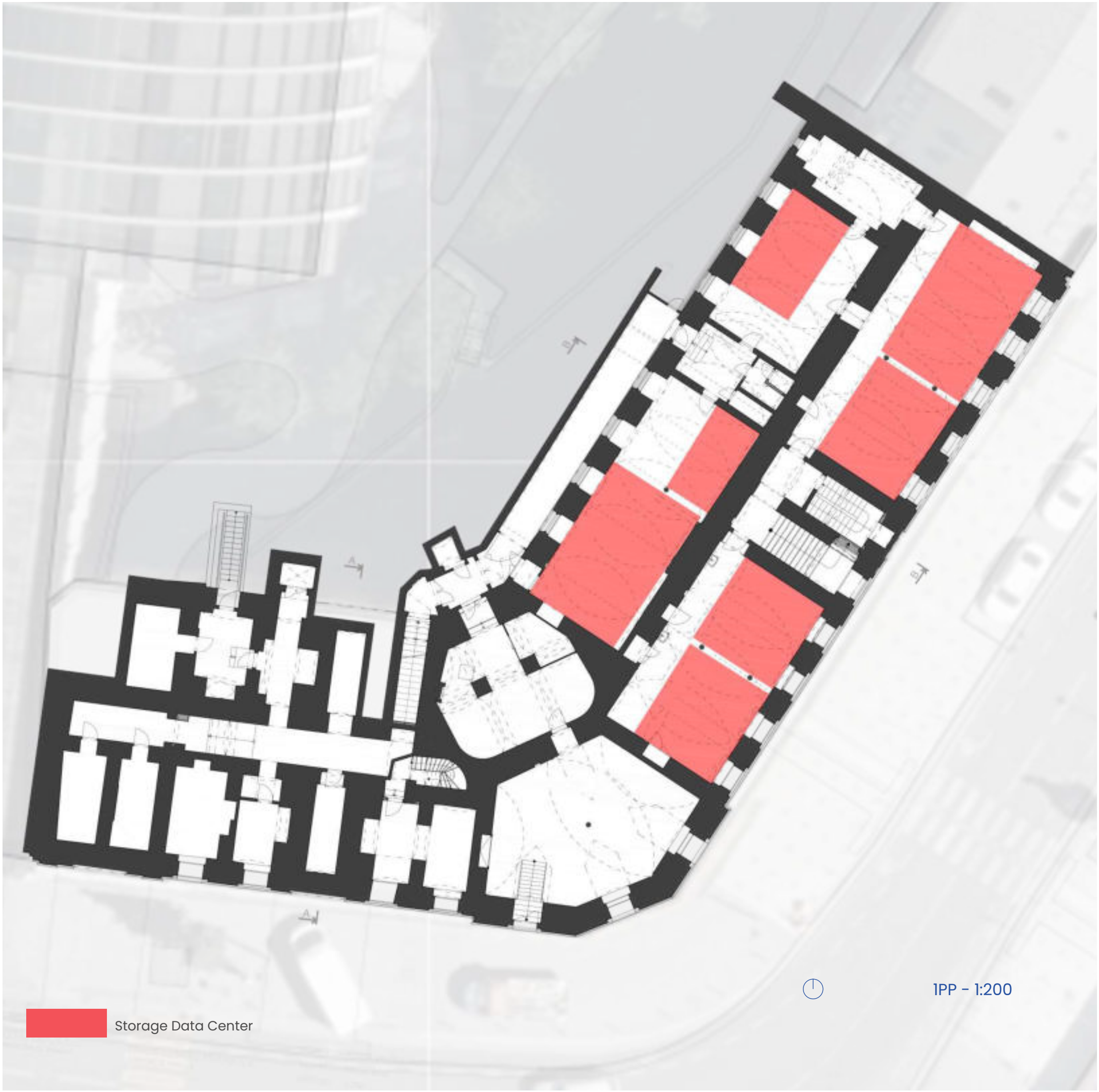
DATA CENTER

STORAGE DATA CENTER

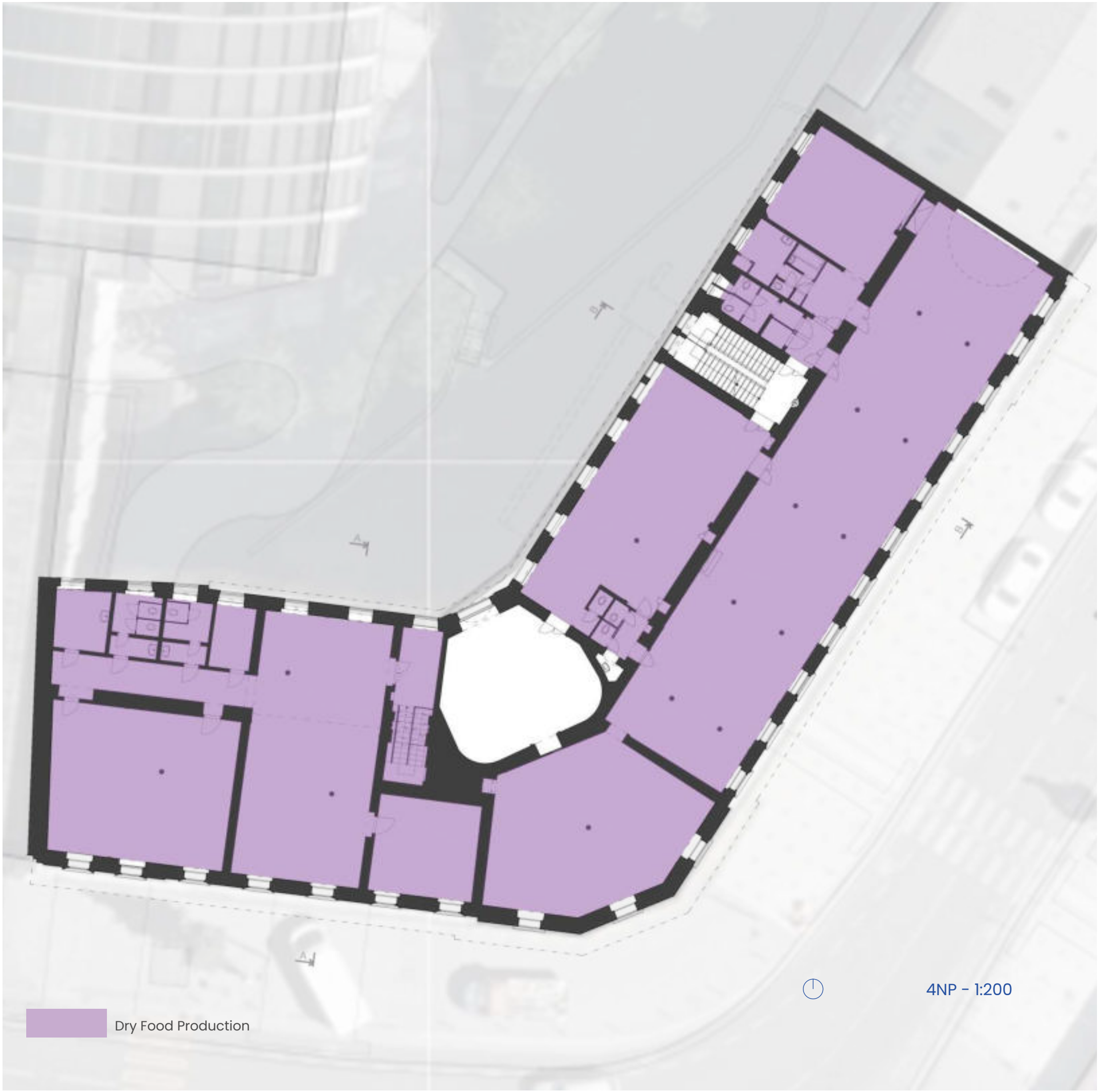
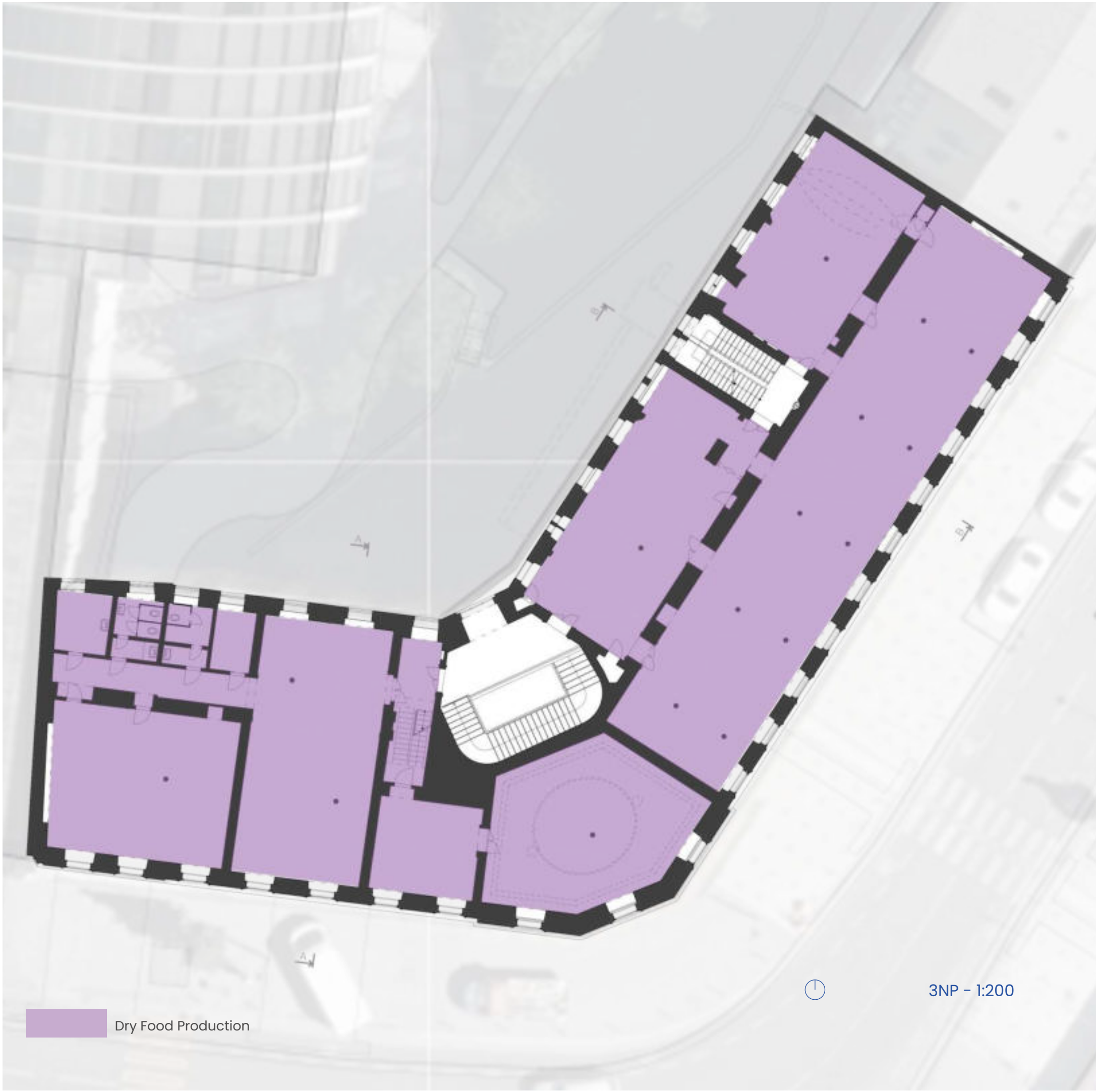
1565 m² P1

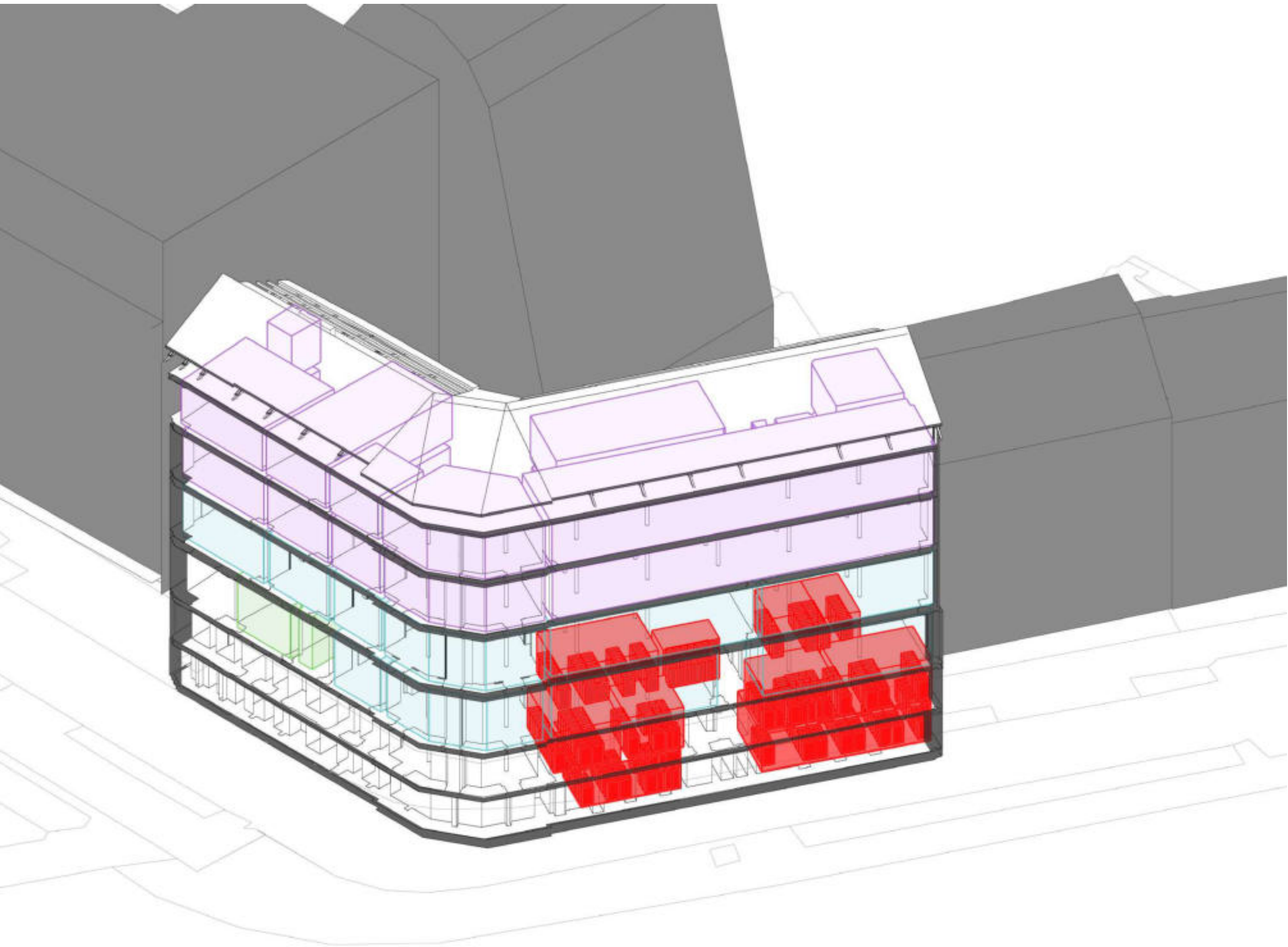
AMOUNT OF SERVERS

300 P1







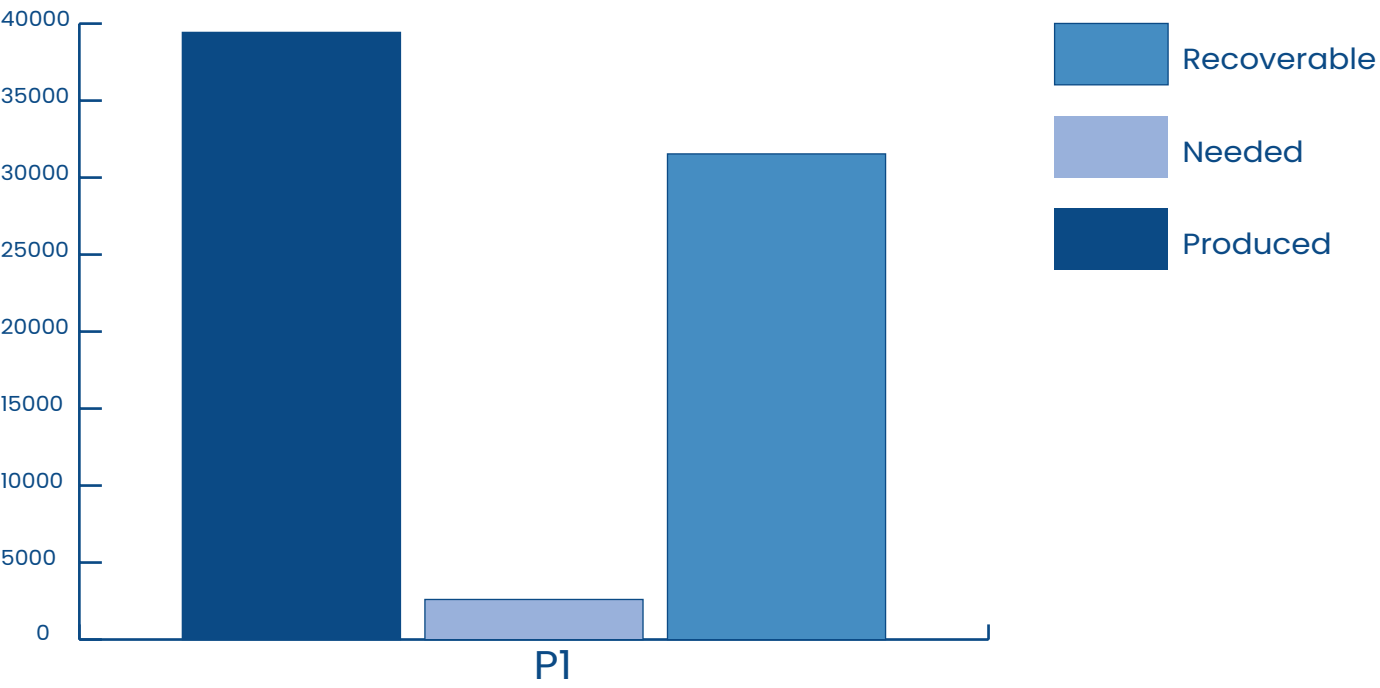


P1 Heating recovery calculation

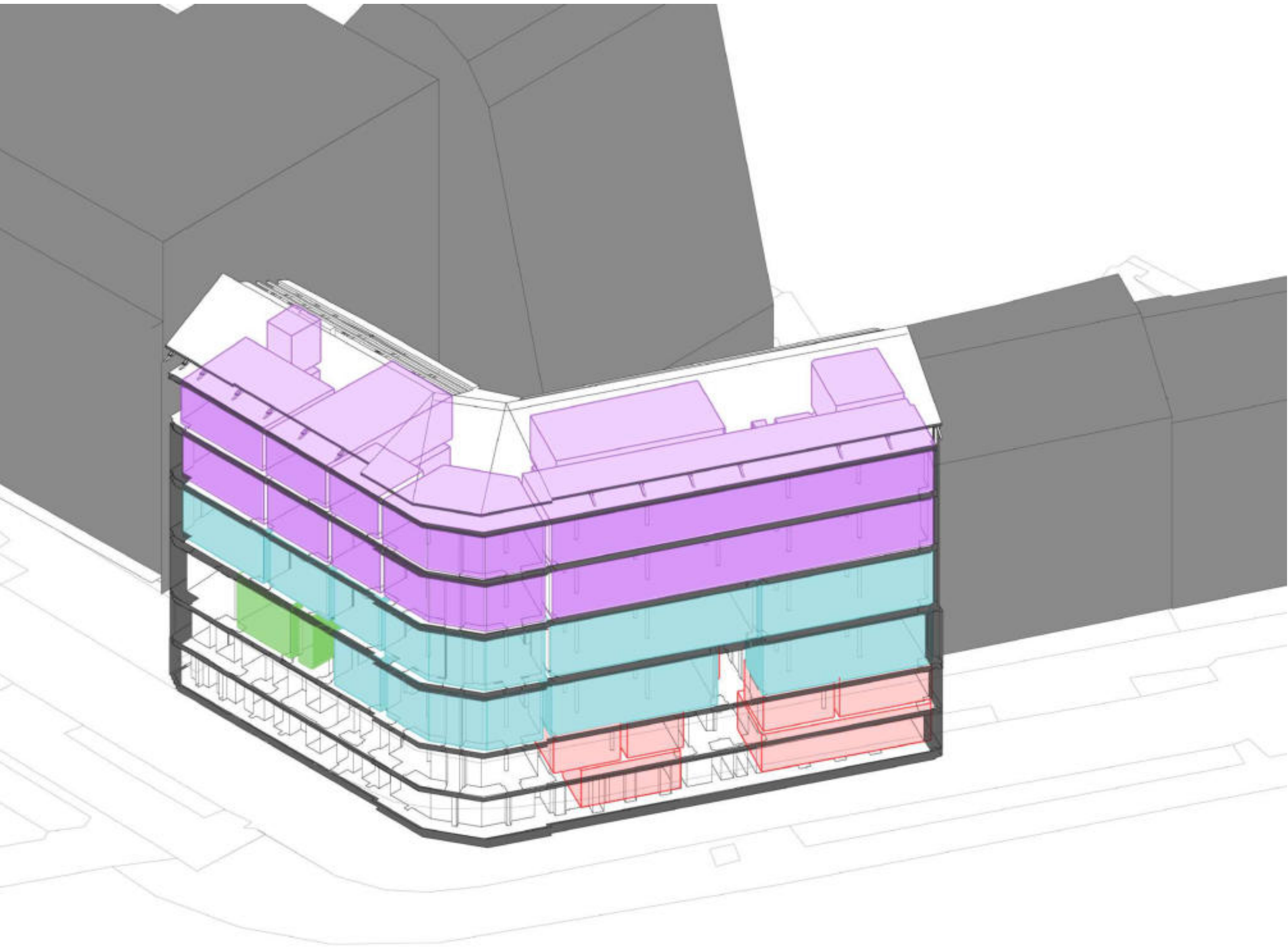
Heat production		
P1	4500 kW	4.5 MW

Heat needed	
P1	2627.95 MWh

Heat recoverable	
P1	31536 MW



Heating recovery		kW/m
Energy consumption		kW
Water consumption		m3
Scalability		m3
Social impact		job/year

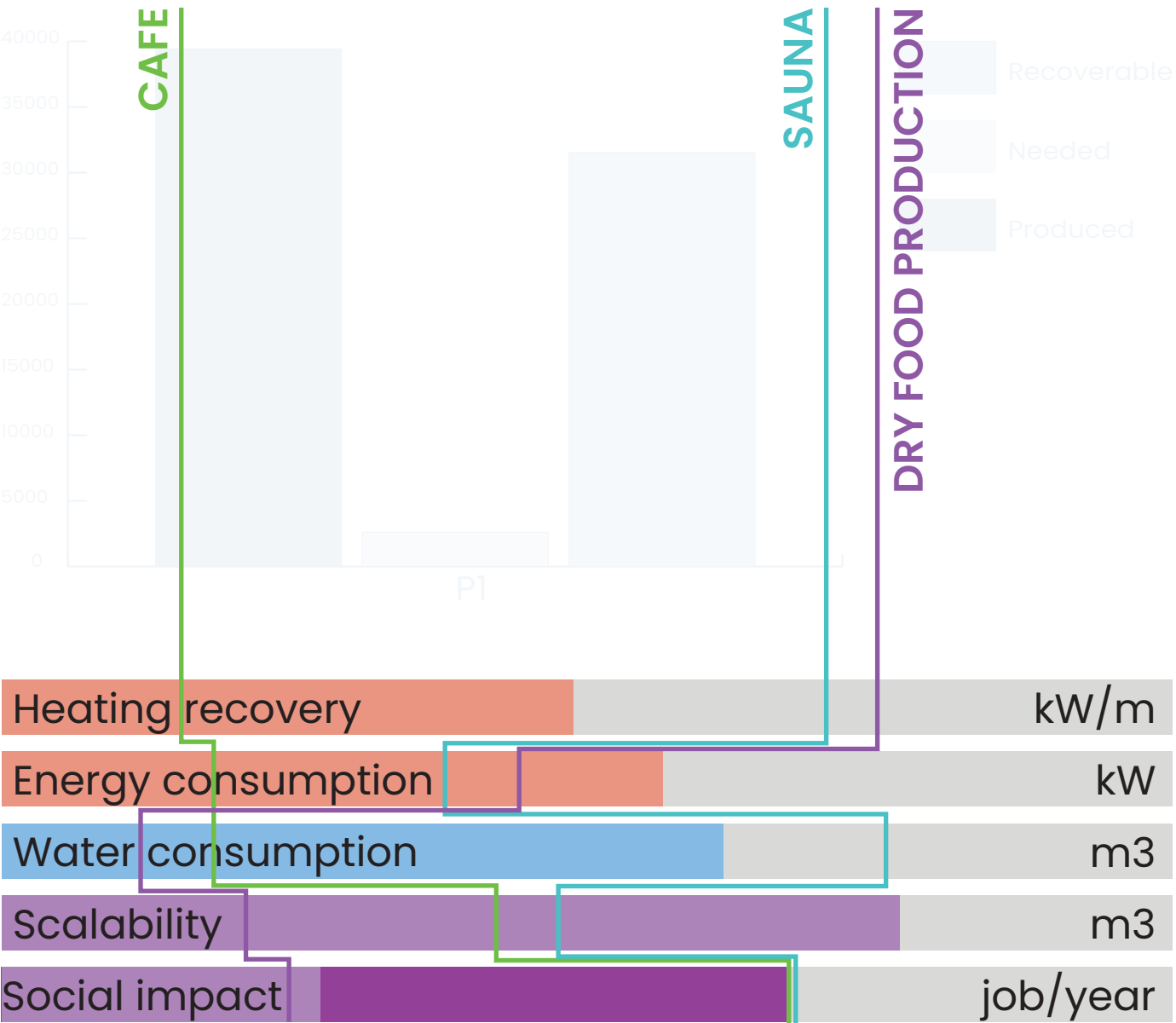


P1 Heating recovery calculation

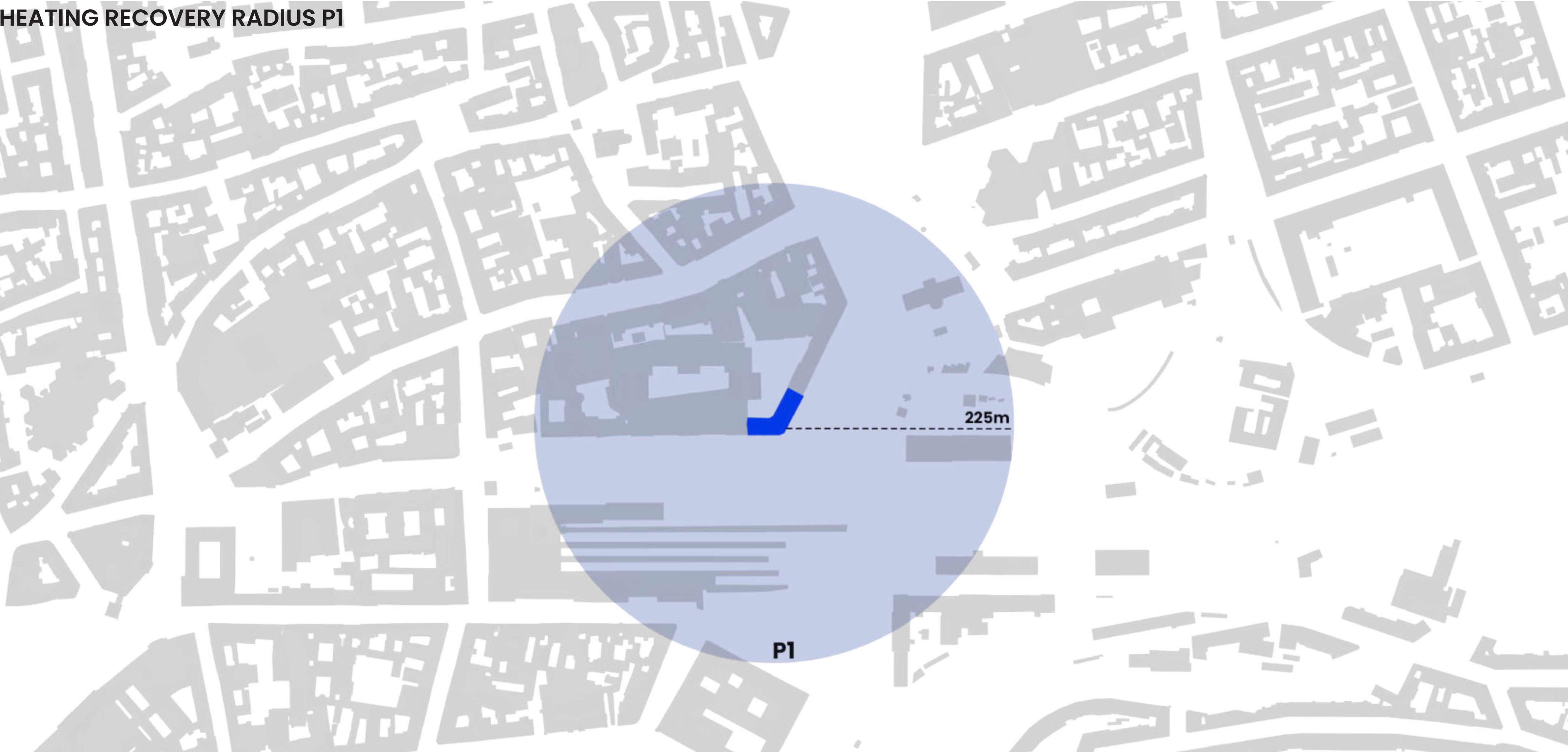
Heat production		
P1	4500 kW	4.5 MW

Heat needed	
P1	2627.95 MWh

Heat recoverable	
P1	31536 MW



HEATING RECOVERY RADIUS P1



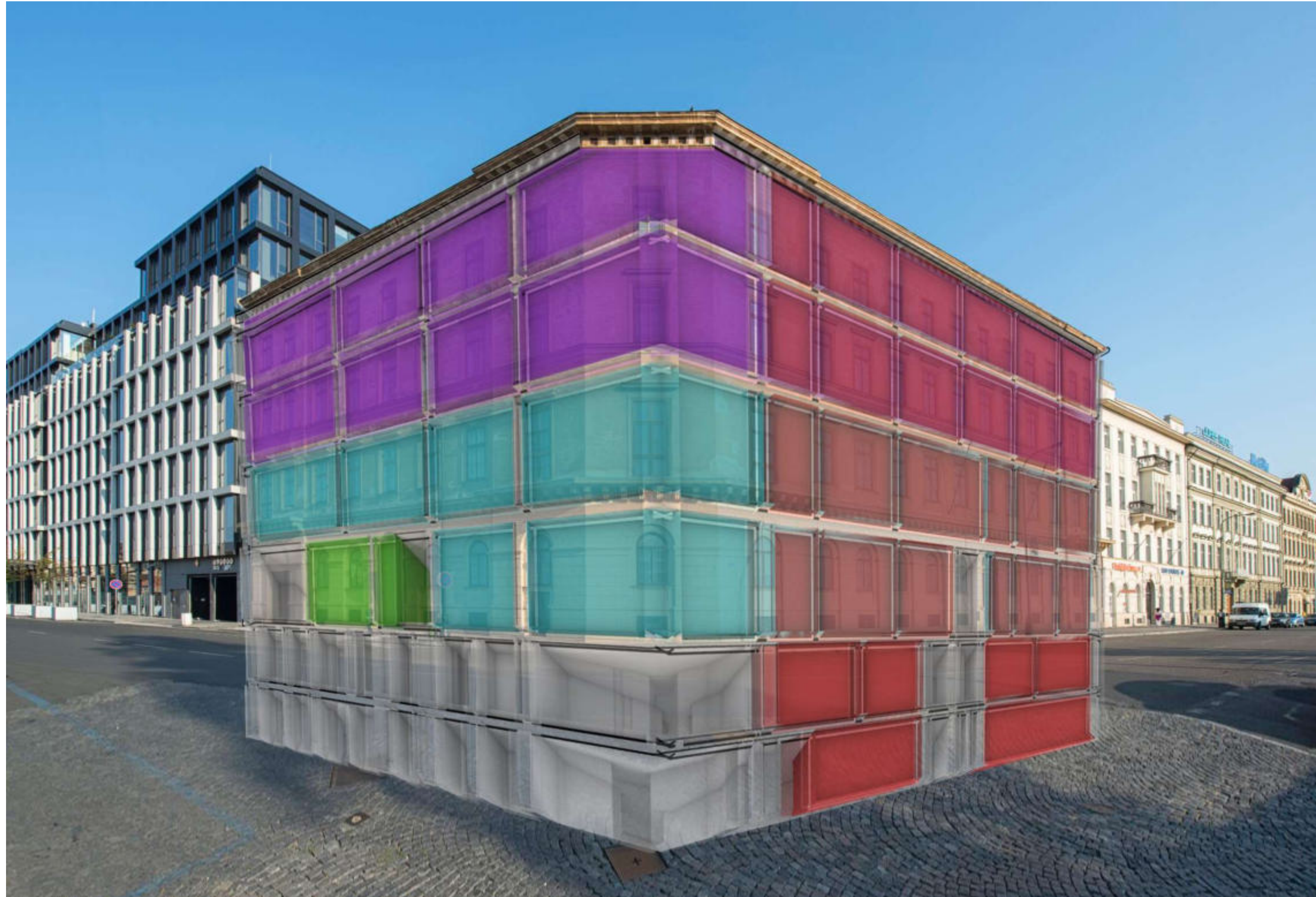
The second phase analyze the potential expansion from the Storage Data Center, meaning the need for more space. By taking 50% of the existing building, the heat produced is increasing as the volume of the mixed-use programs decreases. The building being covered for 100% of the necessary technical infrastructure, it allows for a flexible layout.

The hybrid design is still trying not to interfere with the social impact and the synergy within the structure.

P2

50% MIXED-USE

50% DATA CENTER



MIXED-USE PROGRAMS

DRY FRUITS PRODUCTION

P1	5293 m²
P2	4925 m²

SAUNA/WELLNESS SPACE

P1	4760 m²
P2	3789 m²

CAFE

P1	587 m²
P2	587 m²

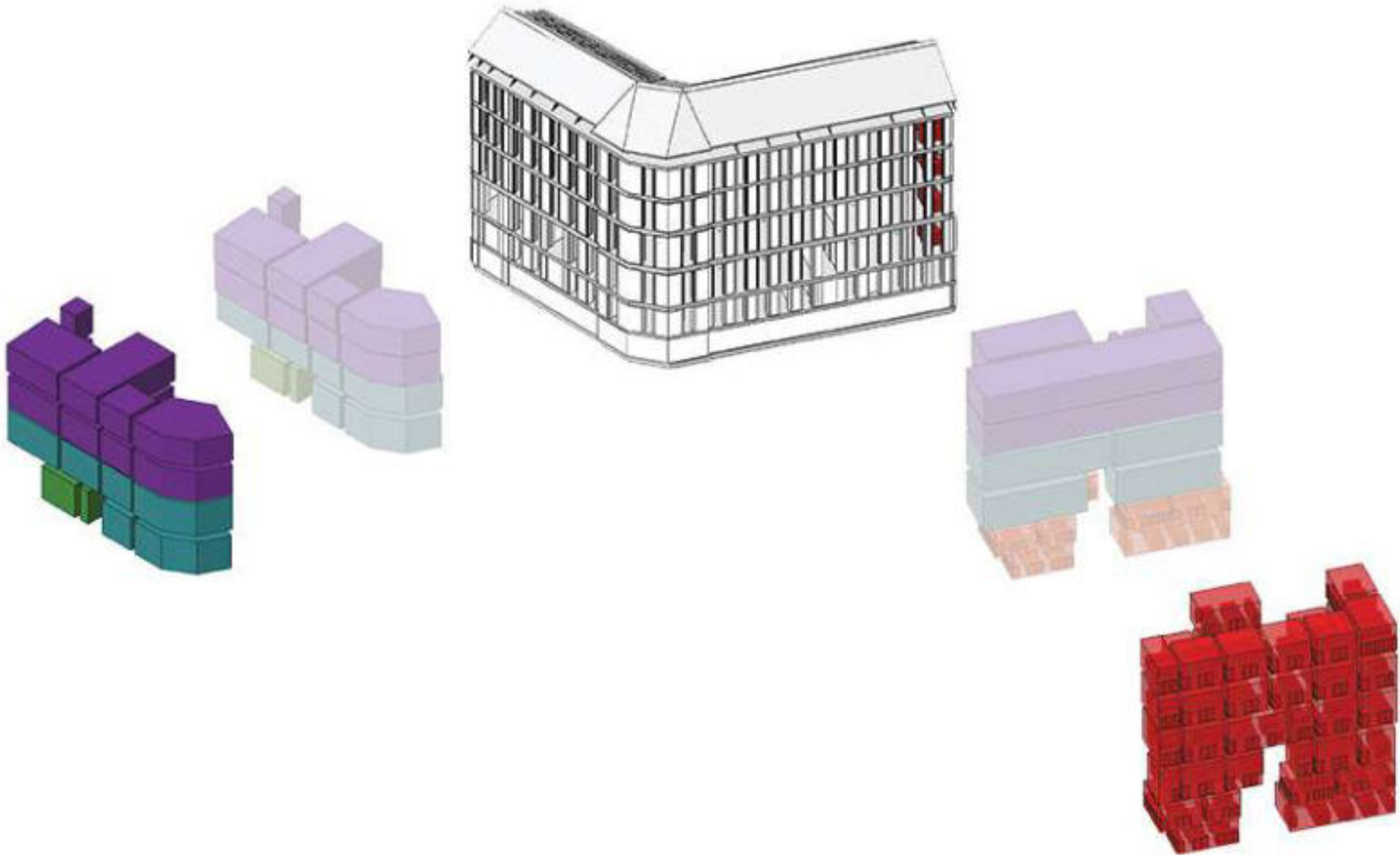
DATA CENTER

STORAGE DATA CENTER

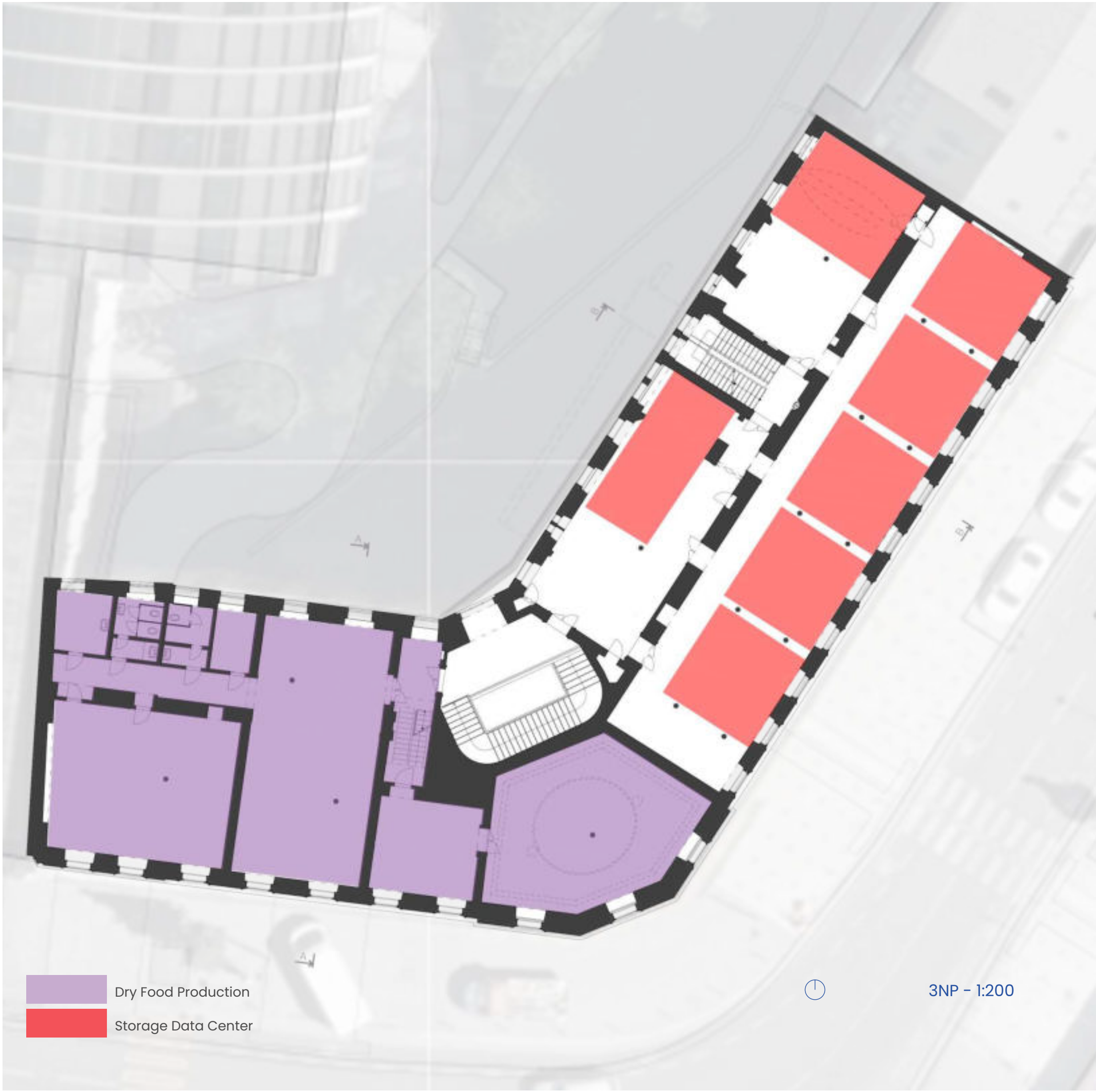
1565 m²	P1
16 703 m²	P2

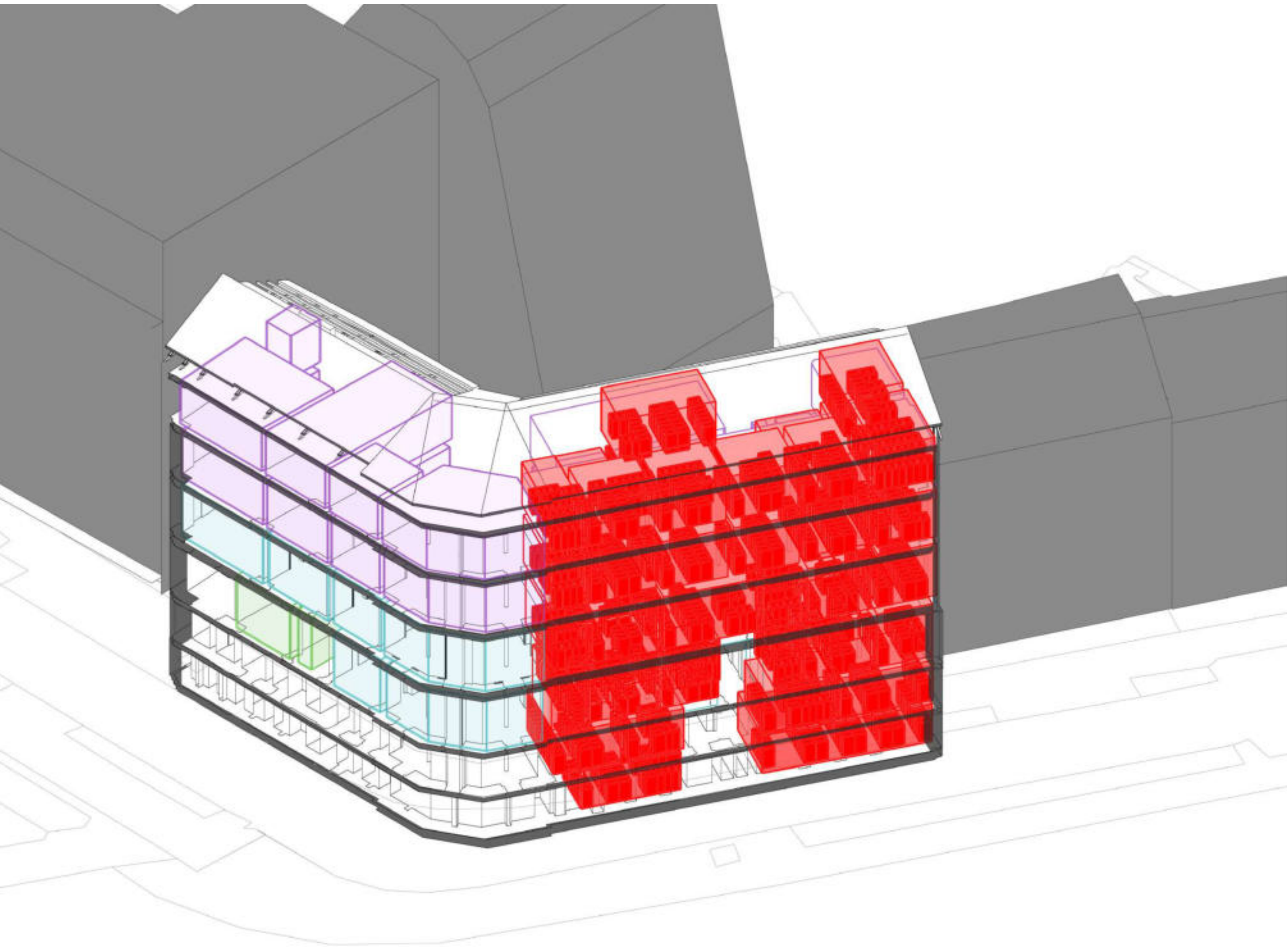
AMOUNT OF SERVERS

300	P1
1150	P2







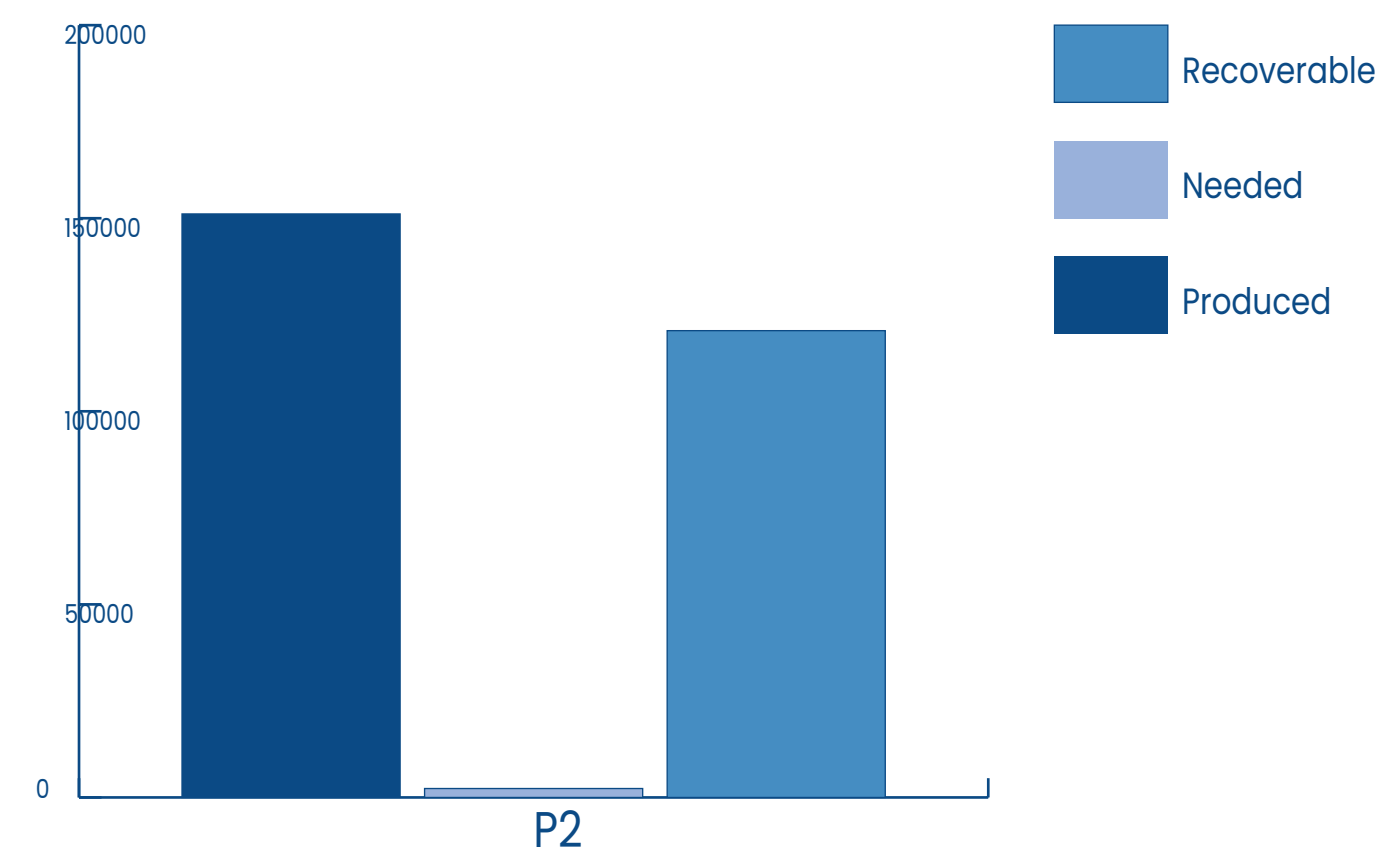


P2 Heating recovery calculation

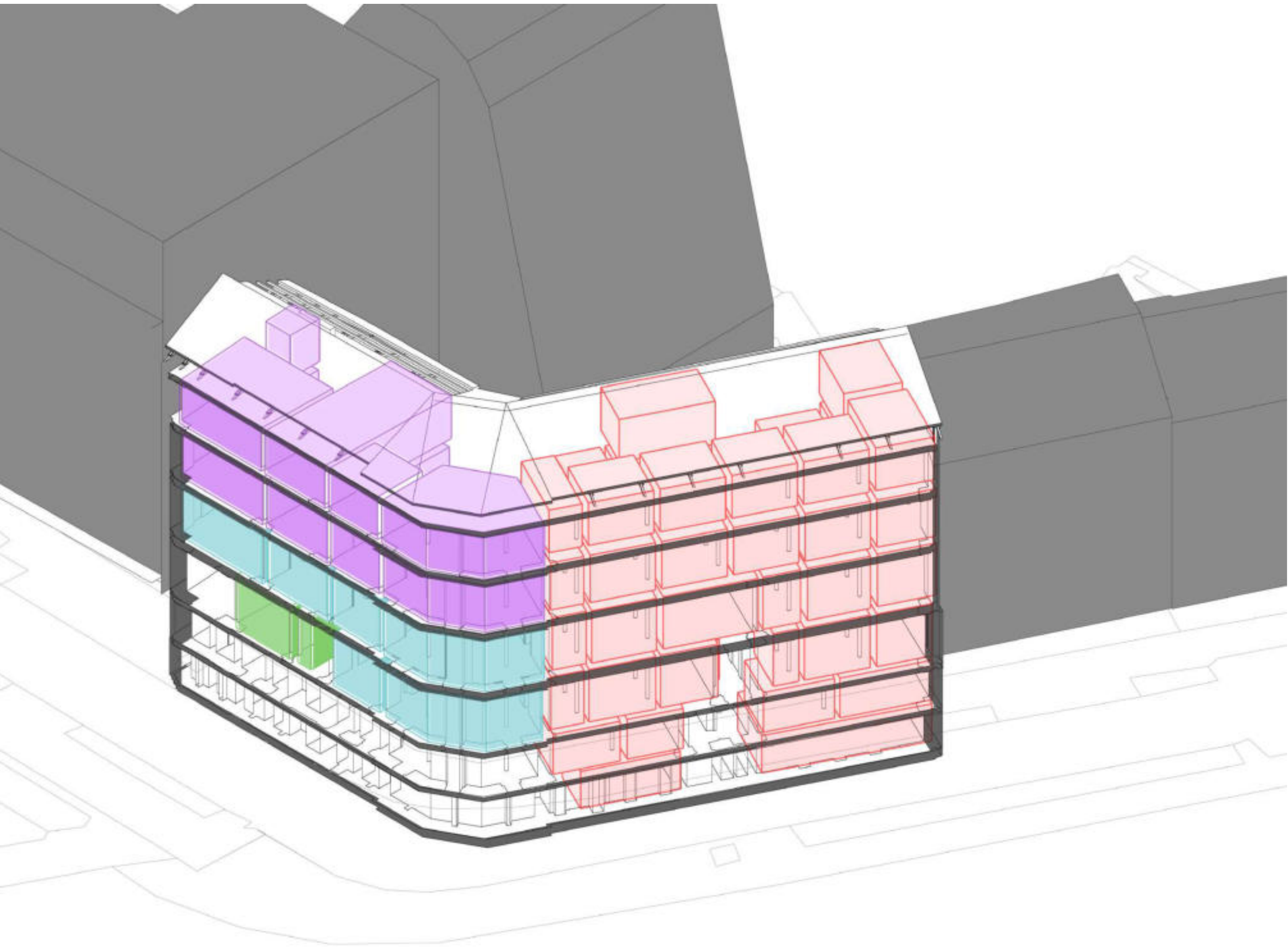
Heat production		
P1	4500 kW	4.5 MW
P2	17250 kW	17.25 MW

Heat needed	
P1	2627.95 MWh
P2	2323.35 MWh

Heat recoverable	
P1	31536 MW
P2	120888 MW



Heating recovery		kW/m
Energy consumption		kW
Water consumption		m3
Scalability		m3
Social impact		job/year



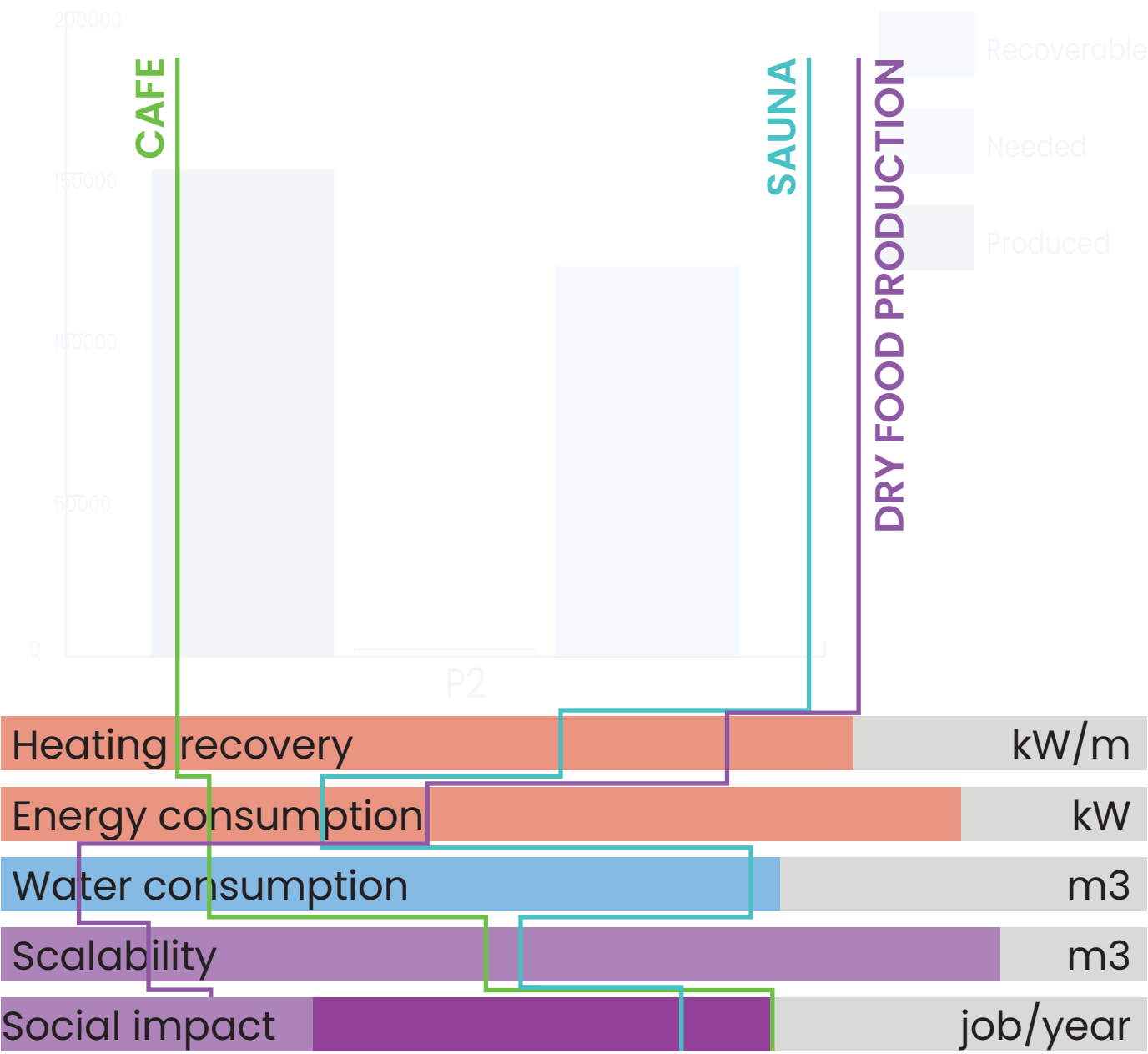
P2

Heating recovery calculation

Heat production		
P1	4500 kW	4.5 MW
P2	17250 kW	17.25 MW

Heat needed	
P1	2627.95 MWh
P2	2323.35 MWh

Heat recoverable	
P1	31536 MW
P2	120888 MW



HEATING RECOVERY RADIUS P2



The last phase of the project is the most hypothetical of them all. The Storage Data Center is taking 80% of the building, again supported by the technical infrastructure covering the entire structure.

The heat production from the racks becomes so important that the structure is used as a generator for the surrounding buildings.

To keep at least a small positive social impact, the cafe remains the only program left.

Finally turning the representative building into an technical infrastructure within the city, a Phantom Data Center.

P3
20% MIXED-USE
80% DATA CENTER



MIXED-USE PROGRAMS

DRY FRUITS PRODUCTION

P1	5293 m²
P2	4925 m²
P3	0 m²

SAUNA/ WELLNESS SPACE

P1	4760 m²
P2	3789 m²
P3	0 m²

CAFE

P1	587 m²
P2	587 m²
P3	587 m²

DATA CENTER

STORAGE DATA CENTER

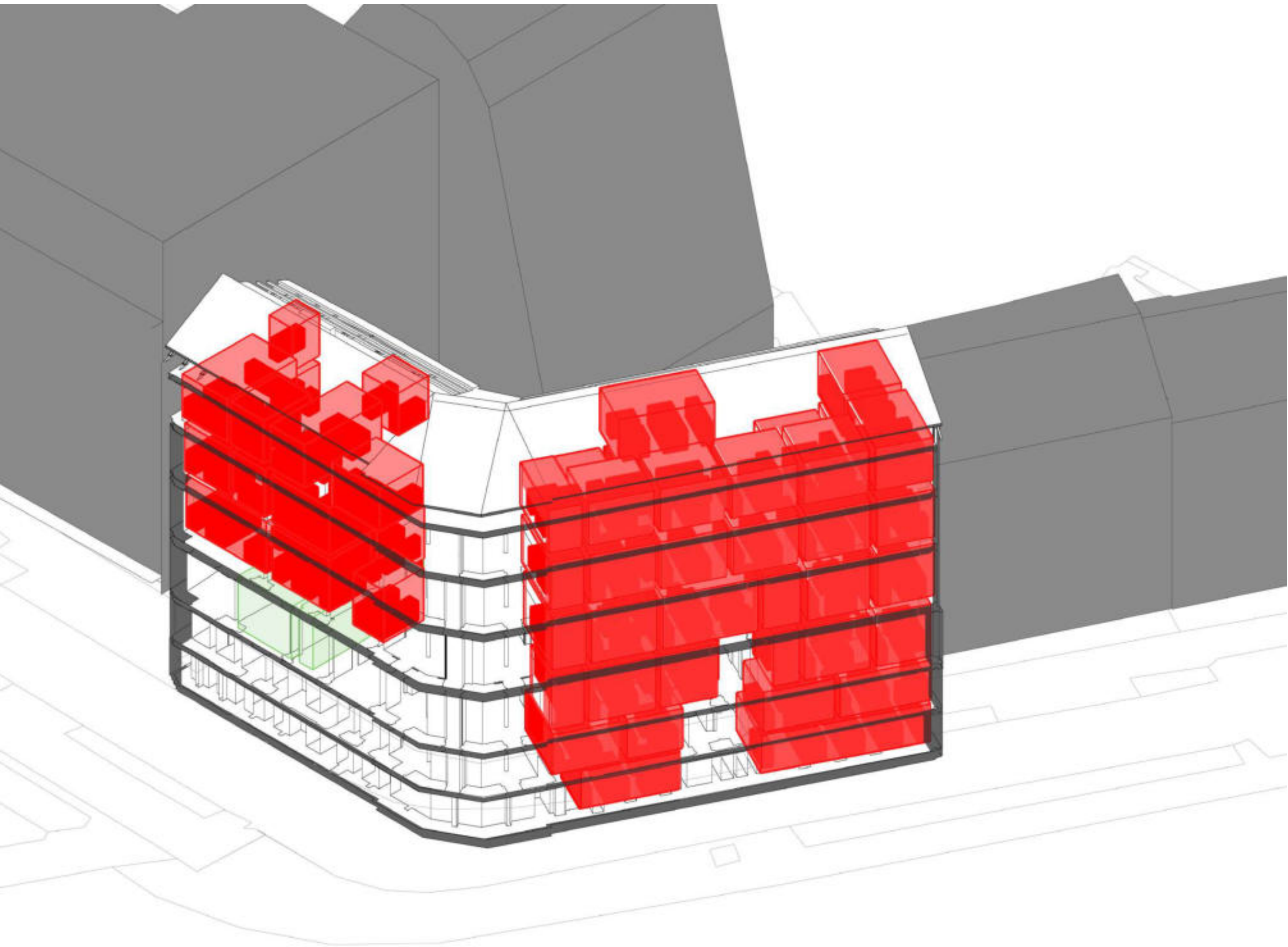
1565 m²	P1
16 703 m²	P2
17 568 m²	P3

AMOUNT OF SERVERS

300	P1
1150	P2
1500	P3



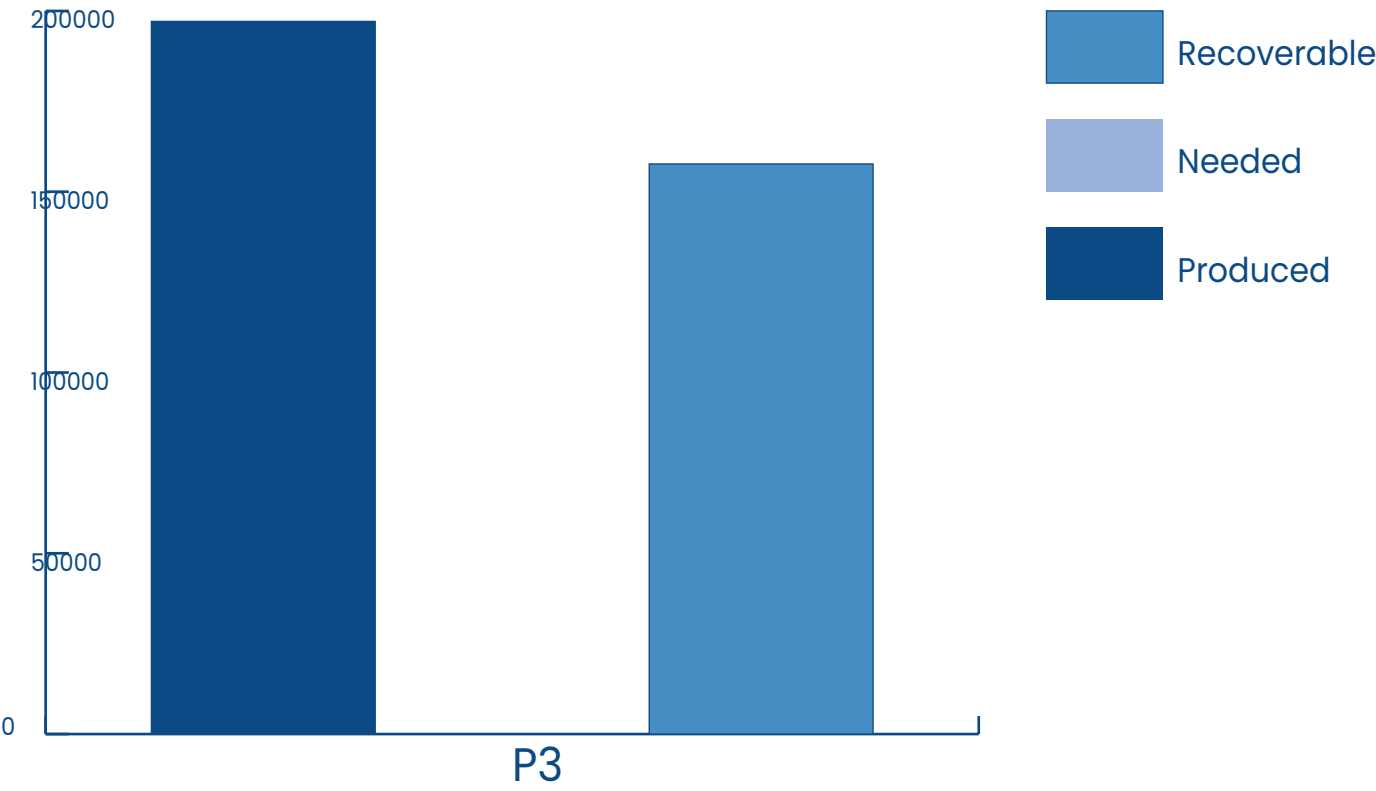




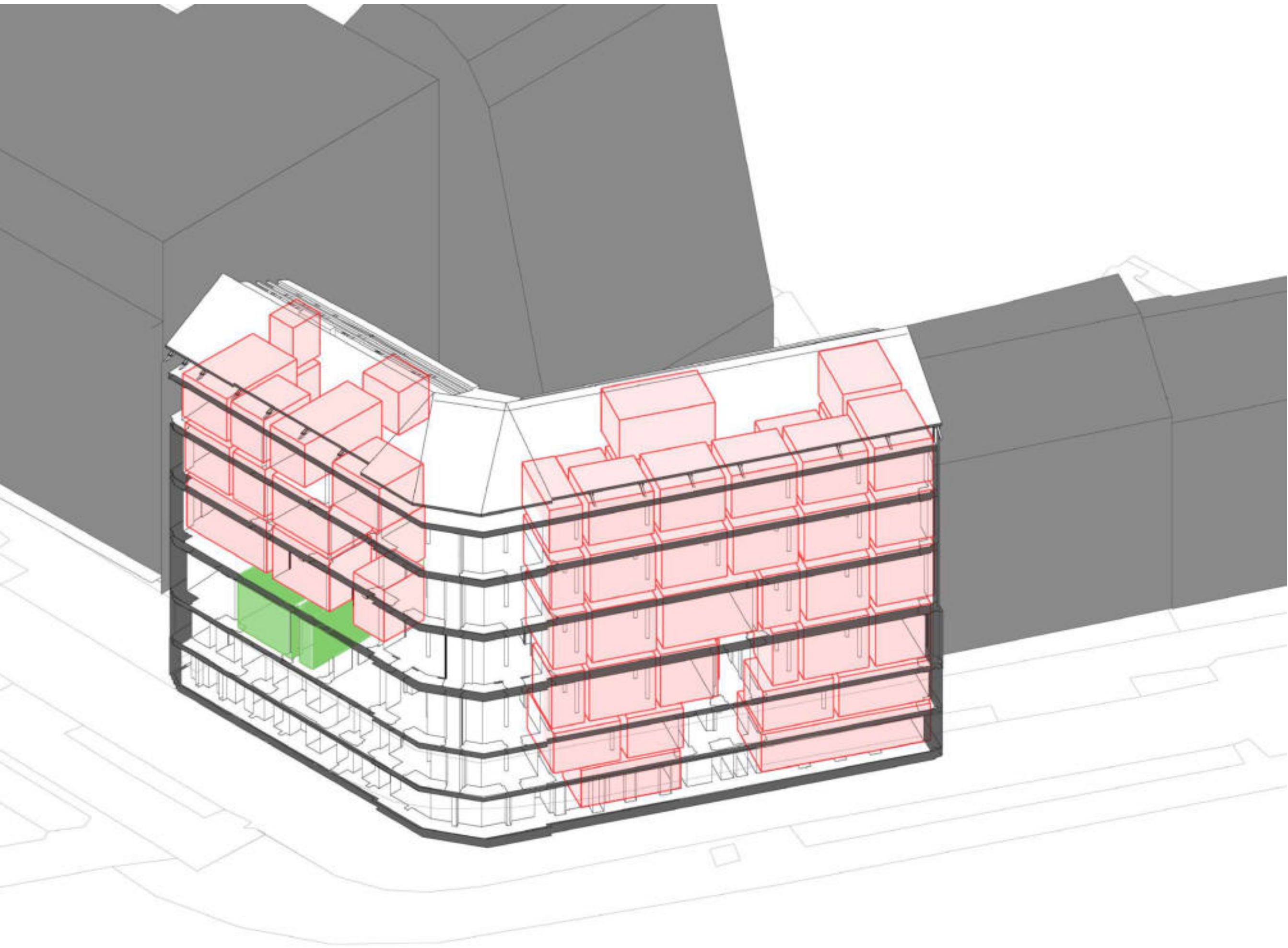
P2

Heating recovery calculation

Heat production		
P1	4500 kW	4.5 MW
P2	4500 kW	4.5 MW
P3	4500 kW	4.5 MW
Heat needed		
P1	4500 kW	
P2	4500 kW	
P3	4500 kW	
Heat recoverable		
P1	4500 kW	
P2	4500 kW	
P3	4500 kW	

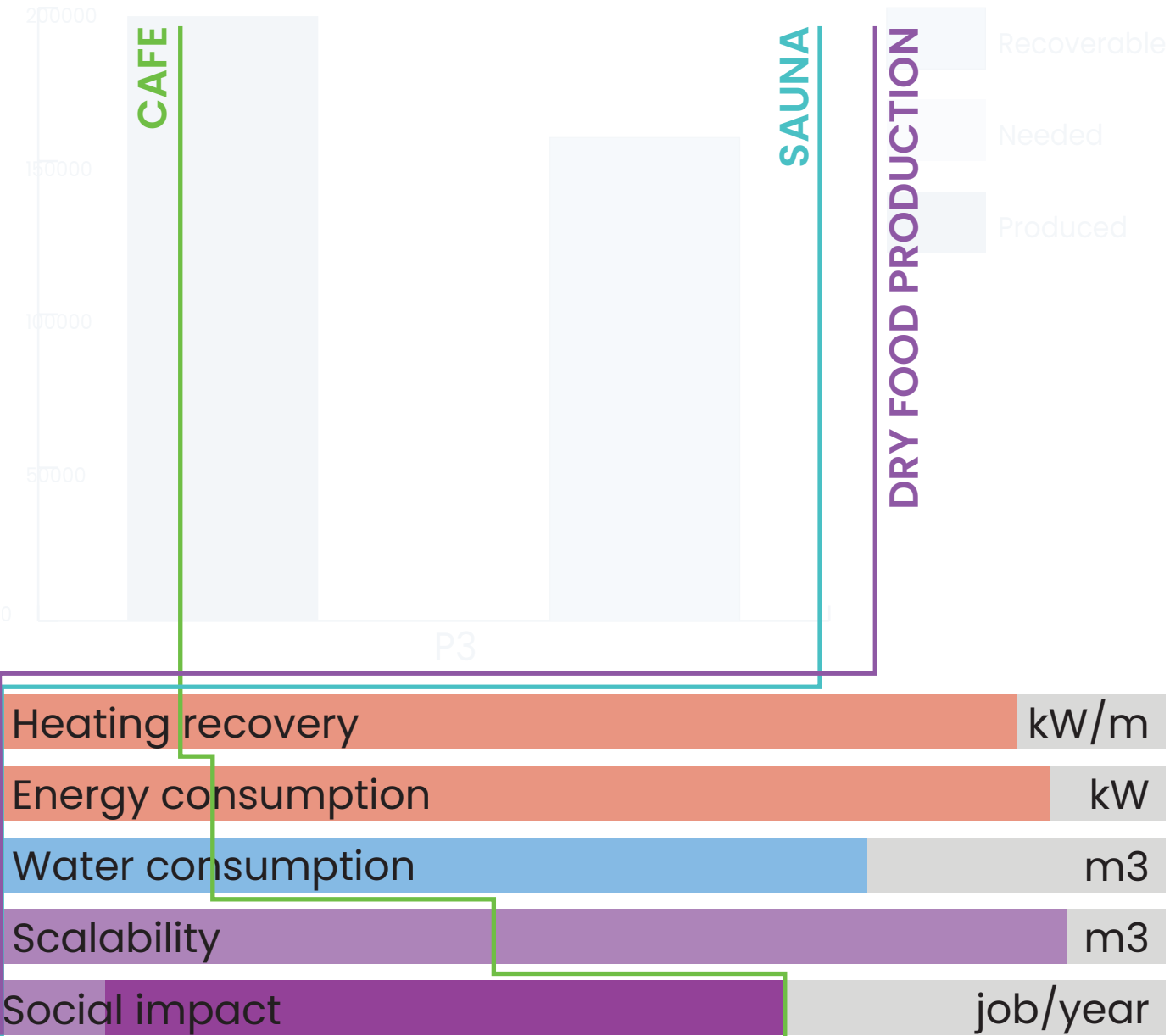


Heating recovery	kW/m
Energy consumption	kW
Water consumption	m3
Scalability	m3
Social impact	job/year



P2 Heating recovery calculation

Heat production		
P1	4500 kW	4.5 MW
P2	4500 kW	4.5 MW
P3	4500 kW	4.5 MW
Heat needed		
P1	4500 kW	
P2	4500 kW	
P3	4500 kW	
Heat recoverable		
P1	4500 kW	
P2	4500 kW	
P3	4500 kW	

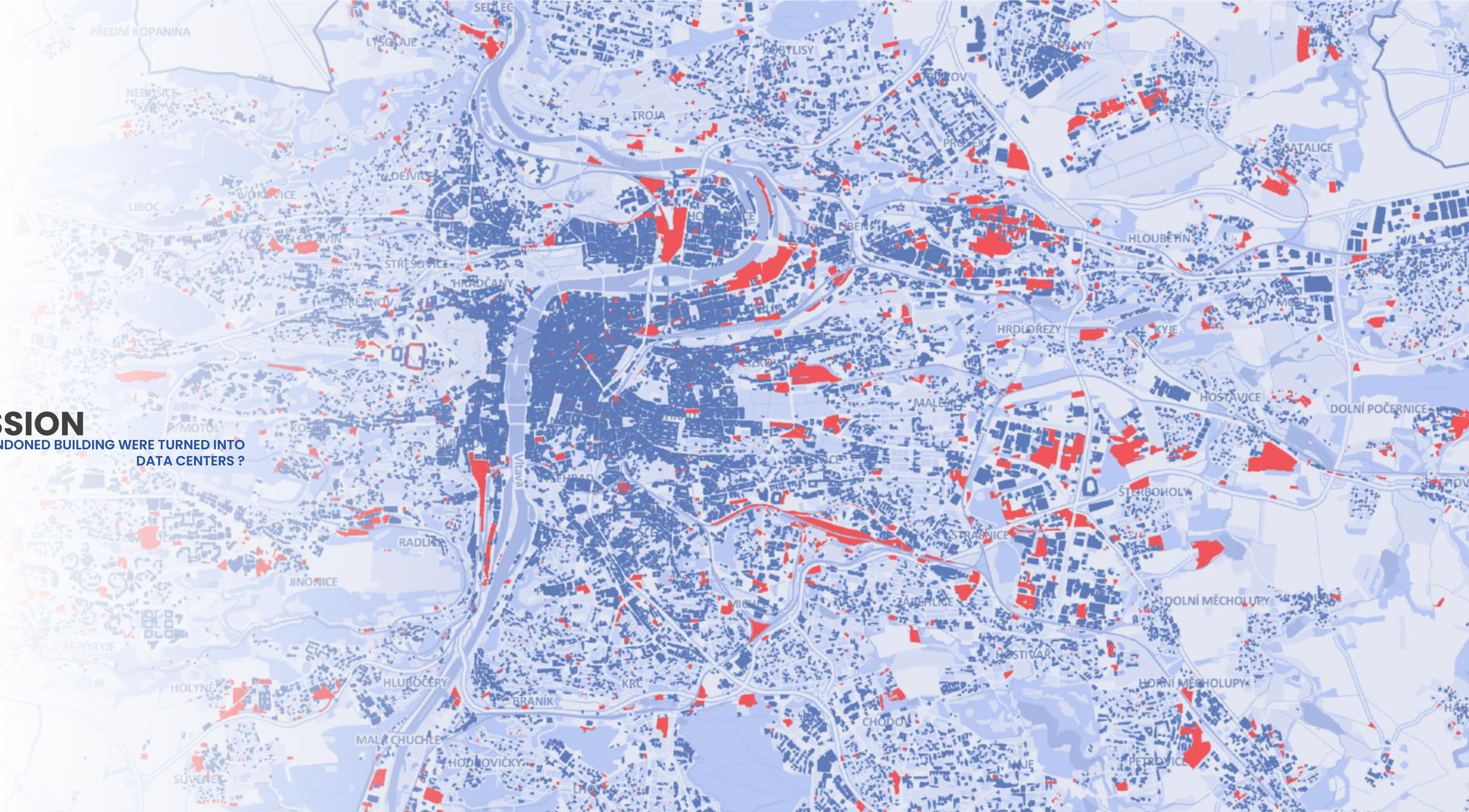


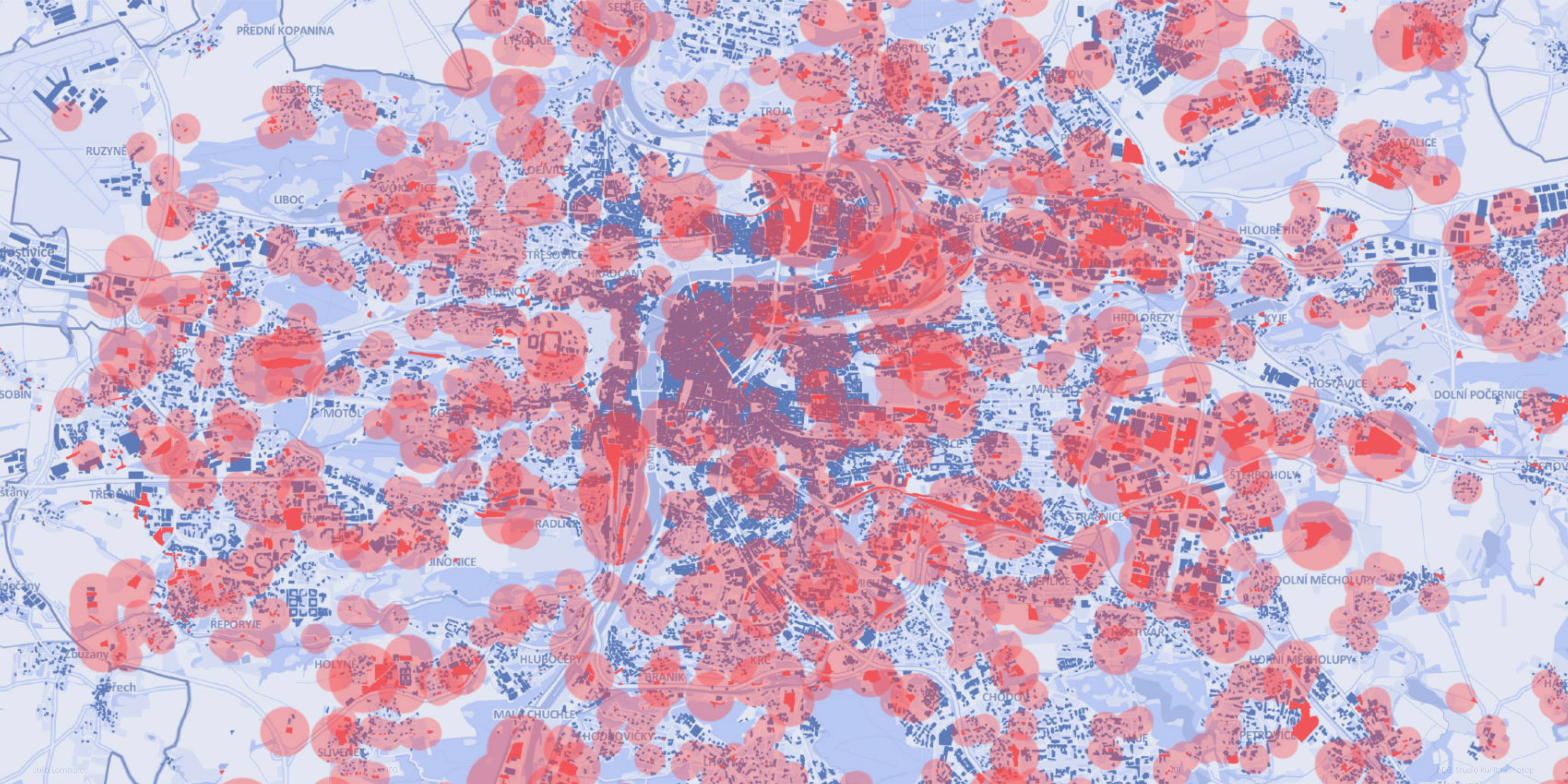
HEATING RECOVERY RADIUS P3



DISCUSSION

WHAT IF ALL ABANDONED BUILDING WERE TURNED INTO
DATA CENTERS ?





RESULT

TECHNICAL REPORT

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1.INTRODUCTION

a. Brief of the project

For the Kurilla–Prokop studio we looked at the different types of warehouses and other buildings in the field of logistics, trying to find solutions on how to make them more appealing and more community-friendly.

b. Goals of the project

The project answers the problematic behind the increasing demand for Data Center and the limitation of space available by using listed abandoned buildings. Because of the impact a Data Center can have on a district, mixed-use programs are also integrated within the structure. The goals are then to analyze the impact a Storage Data Center can have on an abandoned building based on the following Key Performance Indicators: Heating recovery, energy consumption, carbon emissions while thinking about the social impact of it. Meanwhile focusing on the resilience, adaptability and synergy between those very contrasted programs.

c. History of the building

The Desfoursky Palace is used as a case study for this project due to its resemblance with many buildings in Prague. Built by Josef Kranner in 1847, the Neoclassical building was owned by Franz Desfours-Walderode before it got sold by his wife in 1878. It was a representative building, housing administrative and residential programs. Since 1980, the building is listed as cultural monument by the City of Prague.

2. SITE ANALYSIS

a. Context

Located in Florenc in Prague’s New Town, the Desfoursky Palace is accessible through private and public transportation through the extensive network of trams, buses and metro. The surroundings of the Palace is composed of offices, hotels, residential, cultural and educational.

b. Public network analysis – MEP

The building is connected to the mechanical, electrical and plumbing installations. The public network includes power lines, electronic communication, heat pipe, water supply and sewage.

c. Future context

The district is still developing with new projects like the office/residential area called Nova Florenc or the platform over the Masarykovo railway station, improving the connectivity, increasing the amount of people and so the amount of data.

3. PROJECT DATA

a. Existing building

Information about the building				
Building	Desfoursky Palace			
Location	Nov Florenc, Prague			
Built	1847			
Area	6741 m ²			
Stories	2 basement levels + 5 upper floors			
Walls	Height (m)	0.98 to 1.00	0.98	0.98
	Thickness (mm)	100 to 110	100	100
	Thickness (mm)	100	100	100
	Structural system	2011 to 2011	2011	2011
Floor	Masonry slab	0.98 to 1.00	0.98	0.98
	Reinforced concrete slab	0.98	0.98	0.98
	Thickness (mm)	100	100	100
	Thickness (mm)	100	100	100
Roof	Pitched timber			
Materials	Reinforced concrete			

b. Data Center

The Storage Data Center will include an increasing number of servers depending on the phase of the project.

- Phase 1: 300 servers
- Phase 2: 1150 servers
- Phase 3: 1500 servers

The servers are placed in airtight steel temperature-controlled boxes of different scales depending on the placement within the structure.

c. Mixed-use programs

The mixed-use programs were picked based on their heat needs, especially in summer and the potential social impact.

- The Dry food production: Need heat all year long for the drying process while providing to the neighborhood a new form of economy. Acting as a small business to provide to the area local dried fruit, grains and granola for any type of usage.

- For the Sauna/ Wellness space: Interesting for the constant demand for heat all year long and the placement in a very touristic area.

- For the café: Even though a cafe doesn't require that much heat, it's an easy program to incorporate into a district, especially in one constantly evolving.

a. Demolition and preservation strategy

The first phase focuses on preparing the existing structure for the previously named programs, as well as the expanding Storage Data center. A selective demolition of the partition walls and some of the interior load-bearing walls is being carried, excluding the original exterior masonry structure and façade. The existing floor system is also kept original and the need for a raised floor is taken care by temperature-controlled box connecting the technical infrastructure.

b. Programs distribution per phases

The distribution of the programs evolves depending on the phase of the project.

The organization of the mix-used programs depends on their accessibility to the public. The higher floor they're placed at inside the structure, the more private they are. Due to the expansion of the Storage Data Center throughout the phases, those programs' occupancy is decreasing. The café is the only program remaining for all phases and placed on the groundfloor for easy public accessibility.

Mixed-use programs			
	Area (m²)		
Café	PH	1500	NP
	GP	1500	NP
	GP	1500	NP
Storage	PH	4000	1100-2000
Wellness	GP	1500	1100-2000
Spa/Sauna	GP	0	-
Dry Food	PH	1500	2000-4000
Production	GP	1500	2000-4000
	GP	0	-

To prevent further interventions on the structure, the vertical accessibility to the programs is assured by the existing cores including the central marble staircase complimented by new signage.

c. Structural interventions

01.Foundation

The foundation of the building is the two basements made entirely of masonry. The thickness of the walls for the foundation is enough to sustain the additional loads and need no reinforcement. For the project the foundation stays intact, and the basements are being cleaned to welcome the temperature-controlled box for the servers.

02.Vertical load bearing system

The existing building is composed of thick masonry walls, with a timber structure for the existing floor system and to support the roof. The project is reusing the existing external structure entirely while the existing partition and some of the interior masonry walls are being replaced by columns made from recycled steel S235 with a CHS – Circular Hollow Sections profile. The placement of each column is guided

by a 3 by 3 meters grid with a span of 3 by 6 meters between them. The insertion of the steel framework reduces structural dependence on the existing internal load-bearing walls creating a new flexible layout improving the flexibility and organization on the different floors.

03.Horizontal load bearing system

The beams, connected with the columns, are made of the same material recycled stainless steel with an I profile, provides structural efficiency while keeping the existing high ceiling intact. The beams placed perpendicularly to the external load-bearing walls, are spanning between 3 and 6 meters. The new column-beam system is then able to support the increasing loads introduced by the expanding Storage Data Center. The system remains visible to reinforce the contrast between the historical structure and the project.

d. Non-structural system

The Palace was originally constructed without any type of insulation system. The thick envelope of the masonry structure was acting as is due to thermal mass capacity. To assure thermal comfort a layer of internal insulation is being introduced.

a. Materiality

Height (m)		5.1	
Walls	Thickness (mm)	OPRINTOP	1200
		1800	110
		OPRINTOP	650
Structural system (m)		6.5	
Roof		Reinforced timber	
Materials		Masonry - Timber	

The existing building is a thick masonry structure with a timber structure to support the roof. The project is reusing the existing structure as much as possible. The only addition of materials is internal insulation with the following layer composition (from outside to inside).

01.Roof

Existing: Roof slates – ventilated air gap – waterproofing membrane – glass wool thermal insulation – wood paneling
Project: Dense wood wool board – vapor barrier – wood paneling

02.Wall

Existing: Solid bricks – plaster
Project: Vapor retarder – dense wood wool board – vapor barrier – OBS panel – Lime-gypsum plaster

03.Floor

Existing: Timber beam – cross laminated timber – protective layer – cement screed
Project: Vapor barrier – dense wood wool board – parquet

b. HVAC

For the heating of the structure, a heating recovery system connects the Storage Data Center to the mixed-use programs part. The servers are placed in temperature-controlled boxes, with a cold-hot aisles system. Because the number of servers increases depending on the phase, the heat will be then recovered for the adjacent buildings and finally for the district.

The servers from a storage data center need to be stabilized in a temperature range of 10 to 35°C. To ensure that temperature range two cooling systems are used and complimentary.

The first one is the addition of dry coolers fans in between the roof structure. The roof is being split into two parts, depending on their visibility from the street. While the most visible one stays intact, the hidden one is being partly opened for the fans to collect the outside air. The second one is a direct-to-chip liquid cooling integrated in the controlled boxes and directly into the servers. The liquid is being cooled by the dry cooler on the roof and then flows into the servers directly to collect

the heat before being recovered.

The heat from the servers is being recovered and used through the entire building and for the district depending on the phases. The number of servers evolves during the phases and so heat production does too. The heating and cooling system is covering the entire building from the first phase to be able to accommodate any programs to emphasize the idea of adaptability throughout all the phases. Although the capacity of reusable and rejected heat isn't similar for all of them.

For the heating of the structure, a heating recovery system connects the Storage Data Center to the mixed-use programs part. The servers are placed in airtight temperature-controlled boxes, with a cold-hot aisles system. The number of servers is not the only thing evolving. The Storage Data Center's area increase from phase 1 to phase 3, reducing the area used by the mix-used programs and so their heat need.

For each program the heat needed will be assured by the one recovered from the Storage Data Center but not only for the spatial heating, but also for example the drying process of the Dry food production.

Because the number of servers increases, the heat will be then recovered for the adjacent buildings and finally for the district.

The ventilation system covers the entire building just as the heating and cooling system. Because those systems are already in place and focus on the heat recovery and cooling of the Storage Data Center, the ventilation serves for humidity and air quality control with a mechanical air supply and exhaust.

c. Water management

The water supply is connected to the public water network.

After usage, the wastewater will be separated into blackwater and graywater. The blackwater will be directly going into the sewage system. The rainwater will be collected from the open part of the roof and stored in the basement. The graywater will be stored in the basement, combined with the rainwater and filtered to be recycled.

d. Electricity & lighting

The building is relying partly on the public electrical network combined with tiles solar panels especially made for protected buildings. The excess energy from the tiles solar panels get stored in case of energy outage.

e. Internet

For the Internet the building is connected to a fiber optic cable that is part of the public network.

f. Fire safety

The thick masonry structure is highly resistant against fire – category F1 or F2/ S1 or S2. The staircases distributing the entire building are acting as emergency exits. In addition, a system of smoke detectors with fire alarms as well as sprinklers are present in every room.

6. Conclusion

In conclusion, the project explores the integration of a data center in the urban fabric by reusing an abandoned building. A restoration proposal for the reduction of embodied carbon and bringing new life to the site. The project is taking place not only in architecture scale but also answering urban challenges. With the key aspect being the recovery of the heat from the data center to support functions like the sauna/wellness space, the dry food production and the café, as well as the district around it. Overall, the Desfoursky Palace is turned into a hybrid multifunctional space, highlighting that data centers can be rethought as part of the city rather than separate from it.

7. References

1. B01 :: Florenc 21
2. Modernizace a dostavba ŽST Praha Masarykovo nádraží | Jakub Cigler Architekti
3. svk_tradicni-stavitelstvi_2022.pdf
4. Desfourský palác - Památkový Katalog
5. Data Center Handbook - Hwaiyu Geng - Google Books
6. <https://zpesystems.com/data-center-temperature-and-humidity-best-practice-zs/>
7. Energy Audit Report for Aman Cafeteria | PDF | Kilowatt Hour | Lighting
8. <https://chariotenergy.com/blog/coffee-shop-electricity-consumption-chariot-energy-texas/>
9. Energy consumption in commercial buildings: challenges and obligations
10. Experimental research and energy saving analysis of an integrated data center cooling and waste heat recovery system - ScienceDirect
11. Solar Roof Tiles 2026 | Costs & Benefits Explained
12. Sustainable_DataCenter_Whitepaper_Digital.pdf

